Multichannel quantum defect theory treatment of triplet gerade and ungerade *d*-symmetry levels of H_2 and its isotopomers

S.C. Ross, Ch. Jungen, and A. Matzkin

Abstract: This work presents a systematic multichannel quantum defect theory (MQDT) analysis of the triplet *d*-symmetry levels of the hydrogen molecule. First, a new compilation of the best available experimental term values for these levels was prepared. Second, *R*-dependent quantum defect matrices for the $^3\Pi_u$, $^3\Pi_g$, and $^3\Delta_g$ states of H_2 were obtained from ab initio potential-energy curves and used in an ab initio MQDT calculation of all known triplet *d*-symmetry rovibronic levels of H_2, HD, and D_2. For a few of these levels previous ab initio calculations have been reported. The agreement currently obtained is generally significantly better than that in previous work. Finally, the quantum defect matrices are used to calculate the electronic transition moments $^3\Pi_u \rightarrow {}^3\Pi_g, {}^3\Delta_g$ as functions of energy and internuclear distance, *R*, for application in a companion article.

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Résumé : Dans le cadre de la théorie des défauts quantiques à plusieurs voies (MQDT), nous présentons une analyse systématique des niveaux triplets à symétrie *d* de la molécule d'hydrogène. Nous compilons d'abord les meilleures valeurs expérimentales disponibles pour ces niveaux. Par la suite, nous obtenons les matrices de défaut quantique (dépendent de *R*) pour les états $^3\Pi_u$, $^3\Pi_g$ et $^3\Delta_g$ de H_2 à partir de surfaces d'énergie potentielle obtenues ab initio et les utilisons dans un calcul MQDT aussi ab initio de tous les niveaux vibroniques connus à symétrie triplet *d* des molécules H_2, HD et D_2. La littérature rapporte des calculs ab initio pour quelques'uns de ces niveaux. L'accord obtenu ici est généralement significativement meilleur que dans les travaux antérieurs. Finalement, nous utilisons les matrices de défaut quantique pour calculer les moments des transitions électroniques $^3\Pi_u \rightarrow {}^3\Pi_g, {}^3\Delta_g$ en fonction de l'énergie et en fonction de la distance internucléaire, *R*, dans le but de les utiliser dans un article accompagnant celui-ci.

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1. Introduction

Gerhard Herzberg once wrote "Throughout my scientific life I have been interested in the spectra of atomic and molecular hydrogen." This particular interest continued from his early observation of the Balmer series in atomic hydrogen and climaxed more than 50 years later in his discovery of triatomic hydrogen, H_3. His interest in diatomic hydrogen, H_2, ranged from the quadrupole transitions in the ground state (the observation of which proved the existence of interstellar H_2), through the excited Rydberg states (leading to ever more accurate determinations of the ionization limit), to the continuum that lies above. In fact GH's published work on H_2 stretches over an astonishing 65 years, from 1927 to 1992.

This paper is also dedicated to Nis Bjerre (Aarhus), who died on February 3, 2001. Nis Bjerre was a well-known laser-physicist who had a long-standing interest in the triplet states of H_2. He is remembered for his scientific gifts, his rigor, and his kindness.

In the present work and a companion article, we apply multichannel quantum defect theory (MQDT) to the triplet d-symmetry levels of H_2. It was MQDT that was used in the determination of the ionization potential of molecular hydrogen. MQDT has proven to be a powerful tool for the quantitative understanding of electronically excited states in molecular hydrogen (refs. 1–3). In our previous work the $n = 2$ and $n = 3$ ab initio potential-energy curves were used to obtain the quantum defect functions. Since then high-quality ab initio potential-energy curves[3] [4], have become available for yet higher triplet states corresponding to principal quantum numbers of the outer electron of $n = 4$ and 5. These new results have allowed us to determine the energy dependence of the $^3\Pi_g$ and $^3\Delta_g$ quantum defects as well as to incorporate a doubly excited channel in the treatment of the $^3\Pi_g$ states, thereby enabling inclusion of dissociation in the MQDT calculations in a companion article [5].

In this paper, we present R-dependent quantum defects for the triplet ungerade Π and triplet gerade Π and Δ electronic symmetries of H_2, derived from available ab initio potential-energy curves for $n = 2$ to 5. We then use these quantum defects to calculate:

(i) rovibronic bound level energies of the d-symmetry levels [6], which we then compare with the available experimental levels (d-symmetry levels are those with parity $= -[-1]^N$, where N is the quantum number for the total angular momentum exclusive of nuclear and electron spin), and

(ii) energy and R-dependent electronic transition moments for excitation from the $2p\pi_u c\ ^3\Pi_u$ state into the excited gerade manifold.

In a companion paper [5], we use these quantum defects and transition moments to calculate and interpret both the predissociated [7] and also the simultaneously predissociated and autoionized [8, 9] resonances observed by Bjerre and co-workers. These experimental observations have not previously been accounted for by quantitative calculations.

Ab initio rovibronic level calculations performed by solving the Schrödinger equation directly on the individual ab initio potential-energy curves do *not* yield results in anywhere near as good agreement with experiment as the quality of the curves themselves. This is due to the importance of adiabatic and nonadiabatic effects in excited states. As shall be seen below the ab initio quantum defect treatment outlined in ref. 2 accounts for the bulk of these effects and gives significantly better results for the energies of the rovibronic levels.

2. Quantum defects

Figure 1 displays the ab initio potential-energy curves and Fig. 2 the derived quantum defect functions for $^3\Pi_u$ (Figs. 1*a* and 2*a*; see footnote 3), $^3\Pi_g$ (Figs. 1*b* and 2*b*; see footnote 3), and $^3\Delta_g$ (Figs. 1*c* and

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[3] W. Kołos. Private communication. 1995.

Fig. 1. Ab initio Born–Oppenheimer potential-energy functions used to determine the quantum defects. Horizontal lines show the energies of the $N = \Lambda$ levels of Tables 1 and 2. The broken-line potential curves correspond to the $1\sigma_g$ and $1\sigma_u$ states of H_2^+.

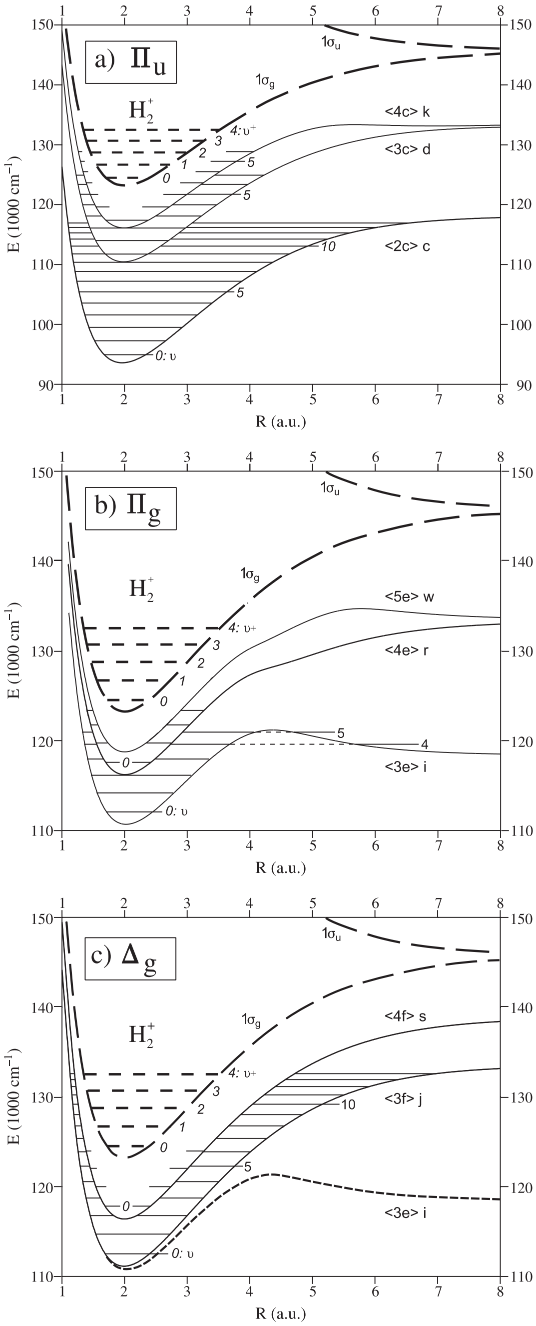
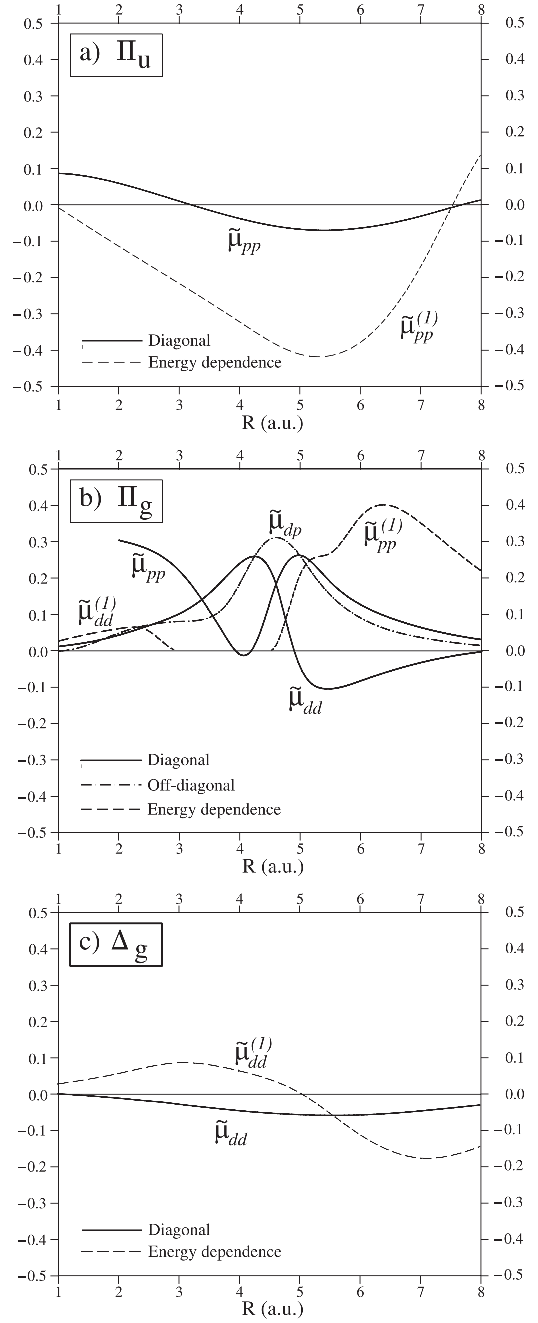


Fig. 2. Quantum defect functions and their energy dependence as obtained and used in this work.



2*c*; ref. 4) symmetry. The quantum defects were extracted from the ab initio potential-energy curves in a way similar to that described in ref. 1. In the present work, however, the quantum defects are defined in a different way from that used in refs. 1–3. In those previous works, the clamped nuclei energies that make up the electronic potential-energy curves were related to the K matrix by the equation (equivalent to eq. 9 of ref. 1)

$$\left| K^{\Lambda} + \frac{\tan \pi \nu}{A(\nu)} \right| = 0 \tag{1}$$

For each value of the projection of the total angular momentum on the molecular axis, Λ, the K matrix is defined in terms of the so-called η-defects (equivalent to eq. 7 of ref. 1)

$$K_{ij}^{\Lambda}(R) = \tan \pi \eta_{ij}^{\Lambda}(R) \tag{2}$$

Here i and j index the electronic channels and $\eta_{ij}^{\Lambda}(R)$ is the R-dependent (and possibly energy-dependent) quantum defect matrix. The quantity ν appearing in (1) is the effective principal quantum number variable (eq. 3 of ref. 1), while $\tan(\pi\nu)$ and $A(\nu)$ refer to the diagonal matrices (eq. 10 of ref. 1)

$$[\tan \pi \nu]_{i,j} = \delta_{ij} \tan \pi \nu_i, \qquad [A(\nu)]_{i,j} = \delta_{ij} A_{\ell_i}(\nu_i) \tag{3}$$

ℓ_i is the orbital angular momentum quantum number of the outer electron in channel i and the function $A_\ell(\nu)$ is given in eq. 5 of ref. 1.

Setting $A(\nu) = 1$ in (1) results in the more usual μ quantum defect, which, in the single channel case, leads to the familiar Rydberg formula. Ham [10] introduced the particular form for $A(\nu)$ given in eq. 5 of ref. 1. Ham's form for $A(\nu)$ tends to avoid the appearance of the unphysical 1*p*, 1*d*, 2*d*, etc. solutions that appear with the usual choice of $A(\nu) = 1$ and the μ-defect. With Ham's choice of $A(\nu)$, (1)–(3) then correspond to the so-called η-defect, as we have written in (2) and used in refs. 1–3. Unfortunately, even the implementation of the η-defect does not guarantee the absence of the unphysical states. In addition, our experience of using the η-defect in H_2 indicates that it has a rather anomalous energy dependence, which can actually exacerbate the problem of the unphysical levels. This energy dependence problem in H_2 when using the η-defect is in contradiction to received opinion [11]. In the present application to $^3\Pi_g$ symmetry it proved necessary to incorporate energy dependence, both in the singly excited and doubly excited channel. The reason for this can be seen in Fig. 1*b* where two doubly excited states (belonging to the $(1\sigma_u)\ np\pi_u$ channel) both play a role in avoided crossings affecting the available ab initio curves, and leading to the long-range form of the <3e> i and <5e> w potentials (see below for a description of the notation used for electronic states). We have, therefore, implemented a new approach, to be discussed in detail elsewhere,[4] which results in the replacement of $\tan(\pi\nu)/A(\nu)$ in (1) with $\tan(\tilde{\beta})$. $\tilde{\beta}$ is a function of ν and is chosen so that $\tilde{\beta}(\nu) = \pi\nu$ for $\nu \geq \ell + 1$. An appropriate choice of $\tilde{\beta}(\nu)$ for $\nu < \ell + 1$ removes the unphysical solutions while simultaneously reducing the energy dependence of the resulting quantum defect and also preserving the extrapolation of the quantum defect into the electronic continuum (which is *not* the case with the η-defect). We denote the quantum defect related to the use of $\tilde{\beta}$ by $\tilde{\mu}$ and the fundamental equation, (1), becomes

$$\left| \tilde{K}^{\Lambda} + \tan \tilde{\beta} \right| = 0, \qquad \text{where now} \qquad \tilde{K}_{ij}^{\Lambda}(R) = \tan \pi \tilde{\mu}_{ij}^{\Lambda}(R) \tag{4}$$

The remaining equations in refs. 1–3 are all similarly modified.

As in refs. 1–3, we treat the energy dependence of the quantum defects as a linear function of the energy of the Rydberg electron, ϵ,

$$\tilde{\mu}_{ij}(R;\epsilon) = \tilde{\mu}_{ij}(R;\epsilon = 0) + \tilde{\mu}_{ij}^{(1)}(R)\epsilon \tag{5}$$

[4] S.C. Ross and Ch. Jungen. Unpublished work.

In what follows, we simply denote the value of $\tilde{\mu}_{ij}(R; \epsilon = 0)$ by $\tilde{\mu}_{ij}(R)$. As shall be seen, energy dependence was only used in the case of diagonal quantum defect functions $\tilde{\mu}_{i=j}(R)$.

In this work, we label electronic states with a combination of Dieke's compact notation and the traditional, or Herzberg, letter label when such exists. Thus, <2c>c represents the $2p\pi_\mathrm{u}\mathrm{c}^3\Pi_\mathrm{u}$ state, labeled by Dieke as 2c. <2c>$^-$c refers specifically to the d-symmetry levels of the <2c>c state. Freund et al. [12] provide a most convenient figure relating Dieke's labeling to the traditional labels (see their Sect. 2 and Fig. 1). This labeling was chosen to correspond to that used in a digital compilation of hydrogen spectra being prepared by one of us (SCR).

Having discussed the general nature of the $\tilde{\mu}$-defects we use in the present work, we now turn our attention to their actual determination and values. We first consider $^3\Pi_\mathrm{u}$ symmetry. Our current interest in this symmetry is due to the fact that the initial state for the experiments described in refs. 8 and 9 is the <2c>c state, the lowest member of the $^3\Pi_\mathrm{u}$ Rydberg series. We have, therefore, treated this symmetry as simply as possible, using a single-channel representation and only including the ab initio potential-energy curves of Kołos[3] for the two lowest states of $^3\Pi_\mathrm{u}$ symmetry, <2c>c and <3c>d ($2p\pi_\mathrm{u}\mathrm{c}^3\Pi_\mathrm{u}$ and $3p\pi_\mathrm{u}\mathrm{d}^3\Pi_\mathrm{u}$), in the fitting of the $\tilde{\mu}$-defect function. The ab initio potential-energy curves are shown in Fig. 1*a*, while the resulting $\tilde{\mu}_{pp}(R)$ quantum function for $^3\Pi_\mathrm{u}$ symmetry and its linear energy dependence, $\tilde{\mu}^{(1)}_{pp}(R)$, are shown in Fig. 2*a*. The <3c>d state was included in the fitting to permit the determination of the energy dependence of the $\tilde{\mu}$-defect, thereby allowing for more accurate accounting of the non-Born–Oppenheimer interactions with the higher lying states. An MQDT study of the higher members of the $^3\Pi_\mathrm{u}$ series would be best performed with the inclusion of the available <4c>k state.[3] It is likely that this will require the inclusion of additional channels into the treatment and we defer this for future work.

As already pointed out the $^3\Pi_\mathrm{g}$ symmetry presents a more challenging problem. The available ab initio curves from Kołos are for the <3e>i, <4e>r, and <5e>w states (the $3d\pi_\mathrm{g}\mathrm{i}^3\Pi_\mathrm{g}$, $4d\pi_\mathrm{g}\mathrm{r}^3\Pi_\mathrm{g}$, and $5d\pi_\mathrm{g}\mathrm{w}^3\Pi_\mathrm{g}$ states), and at small R correspond to the $(1\sigma_\mathrm{g})nd\pi_\mathrm{g}$ channel. These are shown in Fig. 1*b* and present a striking example of the fact that the avoided crossing with a doubly excited state is not a singular occurrence. In this figure, the lowest state, <3e>i, is involved in an avoided crossing with the lowest doubly excited state of $^3\Pi_\mathrm{g}$ symmetry, the $(1\sigma_\mathrm{u})2p\pi_\mathrm{u}$ state. This is the origin of the well-known potential barrier to dissociation that can be seen around $R = 4$ a.u. in the <3e>i state. The higher members of the <ne> (or $(1\sigma_\mathrm{g})nd\pi_\mathrm{g}$) Rydberg series are also involved in avoided crossings with this doubly excited state, as can be seen by their unusual behaviour around the same value of R. What is new in the present MQDT treatment, however, is the need to account for the apparent presence of the next member of the doubly excited series, the $(1\sigma_\mathrm{u})3p\pi_\mathrm{u}$ state, which swoops down from above and is seen because of its avoided crossing with the <5e>w state, which therefore, starting around $R = 6$ a.u., turns down from its rise towards the $n = 4$ dissociation limit and instead ends up at the $n = 3$ limit. To model this behaviour with the MQDT formalism, we are, therefore, forced to use at a minimum two channels:

(1) a $(1\sigma_\mathrm{g})\epsilon d\pi_\mathrm{g}$ channel for the singly excited Rydberg series, which for convenience we will denote as the d-channel, and which is described by the quantum defect $\tilde{\mu}_{dd}(R)$ seen as one of the two heavy continuous curves in Fig. 2*b*, and

(2) a $(1\sigma_\mathrm{u})\epsilon p\pi_\mathrm{u}$ channel for the doubly excited series, which we will denote as the p-channel, and which is described by the quantum defect $\tilde{\mu}_{pp}(R)$ seen as the other heavy continuous curve in Fig. 2*b*.

The MQDT treatment must then also involve the quantum defect accounting for the interaction between these two channels, $\tilde{\mu}_{dp}(R)$, which is shown by the medium weight broken-line–dotted curve in Fig. 2*b*. In our MQDT modelling of this system we have taken the off-diagonal interaction quantum defect to be energy independent, while accounting, inasmuch as is possible, for the energy dependence of the two diagonal defects.

The energy dependence of the $\tilde{\mu}_{dd}(R)$ defect can only be determined over the range of R values for which the ab initio data involve at least two members of the $(1\sigma_g)nd\pi_g$ Rydberg series. Since both the <3e>i and <5e>w states take on a doubly excited character at large R, it is only at small R that we have information on the energy dependence of $\tilde{\mu}_{dd}(R)$. Beyond this range the energy dependence was fixed to zero and smoothly connected with the nonzero values at small R, as can be seen in Fig. 2*b* where $\tilde{\mu}^{(1)}_{dd}(R)$ is shown by one of the light broken-line curves.

In the energy range that we are presently considering the energy dependence of the $\tilde{\mu}_{pp}(R)$ defect (the doubly excited channel) is expected to be quite significant as it involves the first two elements of this series. Again, the energy dependence can only be determined over a certain range, in this case for large values of R. Similarly as for the energy dependence of the $\tilde{\mu}_{dd}(R)$ defect, we smoothly connect the energy dependence of the $\tilde{\mu}_{pp}(R)$ defect to the value zero used in the small R region. $\tilde{\mu}^{(1)}_{pp}(R)$ is shown by the other light broken-line line in Fig. 2*b*.

The strong R-dependence of the $^3\Pi_g$ quantum defect matrix elements and their energy derivatives that is evident in Fig. 2*b* contrasts with the more regular looking curves used for the $^1\Sigma_g^+$ symmetry in refs. 1–3. Other than the fact that forcing the energy dependence functions to zero in certain regions automatically results in them having a rather strong R-dependence, two factors come into play to explain the strong R-dependence seen in Fig. 2*b*. The first factor, of a technical nature, results from the fact that, given the available ab initio data, we are forced to artificially limit the treatment of the $^3\Pi_g$ symmetry to two channels. This is one less than was the case in the treatment of the $^1\Sigma_g^+$ symmetry in ref. 1, meaning that quantum defect matrix used here has only three independent elements, rather than the six independent elements in ref. 1. Thus, fewer free parameters were available to model this system. As a result all of the elements of the quantum defect matrix were fitted *as functions of R* over a range of R values, unlike the case of the treatment of the $^1\Sigma_g^+$ symmetry in the refs. 1–3 series of papers in which an excellent modeling was achievable even though several elements of the quantum defect matrix were there taken as *constant functions of R* over a wide range of R values. The second factor explaining the different behaviour here is more fundamental. We first remember that we are now dealing with two *series* of avoided crossings, which, in addition, occur at larger R values than those modeled in ref. 1 for the $^1\Sigma_g^+$ symmetry. In addition to the fact that these are modeled as having the same interaction quantum defect function, since $\tilde{\mu}_{dp}(R)$ is taken as energy independent, our single-centre MQDT expansion simply becomes ever more inadequate at larger R and, therefore, our quantum defect curves take on a progressively more effective nature as R increases. Note, however, that this shortcoming only affects how well the MQDT treatment will account for the non-Born–Oppenheimer effects, the quality of the input Born–Oppenheimer (B–O) potential-energy curves *is still fully preserved in the MQDT treatment*. This is important because it is exactly in the region where our handling of the non-Born–Oppenheimer effects becomes less precise that these effects become significantly smaller in any case, because of the fact that the Rydberg electron is much more strongly bound at large R (see Fig. 1*b*), and we are still left with an excellent accounting of the physics of the system: for once the trade-offs work in our favour!

The case of $^3\Delta_g$ symmetry is rather simpler. As shown in Fig. 1*c* ab initio potential-energy curves from Wolniewicz [4] are only available for the lowest two members of the Rydberg series, <3f>j and <4f>s ($3d\pi_g\mathrm{j}^3\Delta_g$ and $4d\pi_g\mathrm{s}^3\Delta_g$) and lead to a simple one channel MQDT modeling with an energy dependence. The resulting $\tilde{\mu}_{dd}(R)$ defect for this symmetry and its energy dependence are shown in Fig. 2*c*. (Figure 1*c* also includes the lowest $^3\Pi_g$ state, <3e>i, as this state is involved in the predissociation of the <*n*f> $^3\Delta_g$ states as discussed in the companion article [5].)

3. Dipole transition moments

We have used the $^3\Pi_u$, $^3\Pi_g$, and $^3\Delta_g$ quantum defects derived in the preceding section to evaluate R- and energy-dependent dipole transition moments for excitation from the $c\,^3\Pi_u$ state to the channels related to the triplet gerade manifold of states. The calculations are based on our extended multichannel Coulomb approximation described in ref. 13. Figure 3 depicts the resulting energy-normalized dipole

Fig. 3. Energy and R dependent transition dipole moments functions for excitation from the <2c>c state to the following channels: (*a*) <2c> $c \rightarrow (1\sigma_g)\epsilon d\pi_g$, (*b*) <2c> $c \rightarrow (1\sigma_u)\epsilon p\pi_u$, and (*c*) <2c> $c \rightarrow (1\sigma_g)\epsilon d\delta_g$.

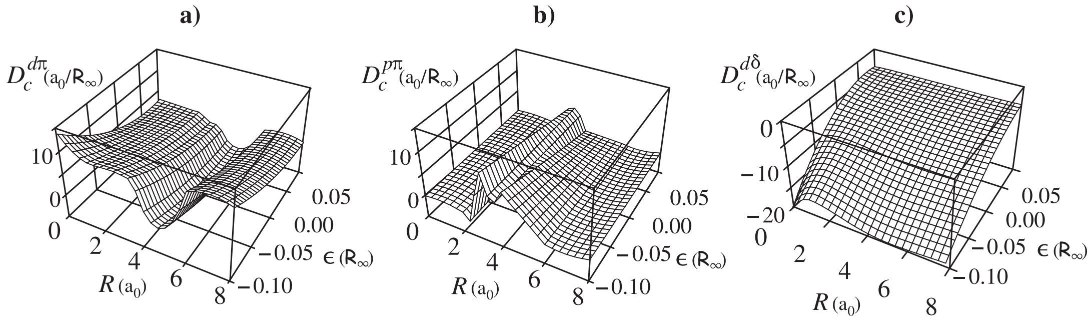
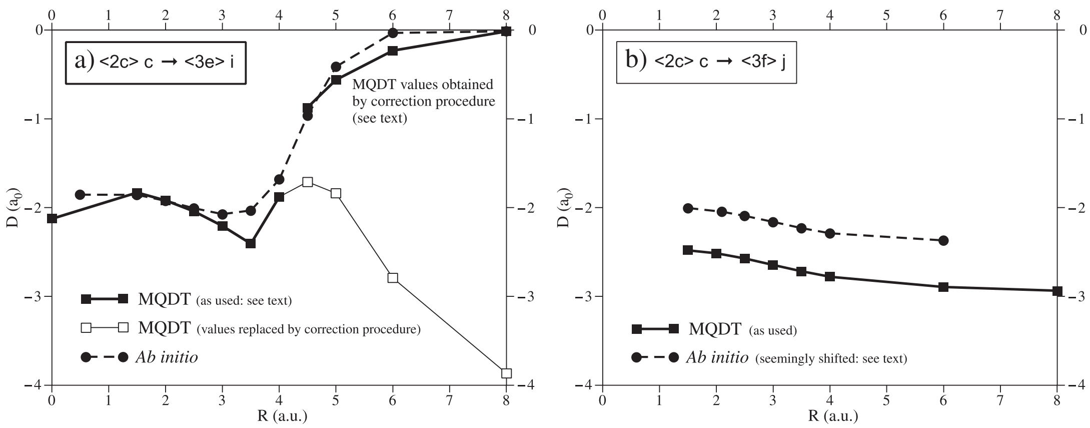


Fig. 4. Comparison of electronic dipole transition moment D (in units of a_0) obtained from MQDT wave functions (squares connected by continuous lines) with ab initio values of ref. 14 (for <2c> $c \rightarrow$<3e>i transition) and ref. 15 (for <2c> $c \rightarrow$<3f>j transition). See text for discussion.



transition moments as continuous functions of R and energy. These transition moments are required in a companion paper [5] where we interpret the photodissociation/photoionization experiments of Bjerre, Helm, and collaborators [7,8,9].

The $c^3\Pi_u \rightarrow i^3\Pi_g$ and $c^3\Pi_u \rightarrow j^3\Delta_g$ discrete–discrete transition moments have previously been evaluated using ab initio methods by Guberman and Dalgarno [14] and by Schins et al. [15]. To verify the quality of our calculated dipole moment functions, we used the quantum defects to construct B–O states in terms of their composed channels and used these in turn to extract B–O-state dependent dipole-moment functions suitable for comparison with the ab initio values. By normalizing our MQDT wave functions to unity we can extract R-dependent moments for these discrete–discrete transitions from the energy and R-dependent ones shown in Fig. 3. These discrete–discrete moments are shown in Fig. 4 where they are compared with the ab initio values. Remember, however, that it is the energy and R-dependent channel dipole moment functions of Fig. 3 that are required for the MQDT calculations. Those obtained from the ab initio calculations are not usable in the MQDT treatment.

Application of the correction procedure [13] to account for the breakdown of the MQDT representation of the wave function at large R (see also the preceding section) results in good agreement between our results for the $c^3\Pi_u \rightarrow i^3\Pi_g$ transition and those in ref. 15 (Fig. 4*a*). In contrast, in the case of the $c^3\Pi_u^- \rightarrow j^3\Delta_g$ transition shown in Fig. 4*b*, our values are systematically about $1.22(\approx [3/2]^{1/2})$ times those of Schins et al. [15] but do agree closely with those recently published by Stolyarov and Child [16].

The transition moment calculated by Schins et al. implies a radiative lifetime of 18.5 ns for the $j^3\Delta_g^-$, $\upsilon = 0$ excited state, in poor agreement with the experimentally determined lifetime of 12.77 ± 0.30 ns measured by Ray and Lafyatis [17]. By simply applying the correction factor of $(1.22)^2$ to the predictions of Schins et al. we find a lifetime of 12.4 ns, in accord both with experiment and with the calculation of Stolyarov and Child.

We note that the $i^3\Pi_g^- - c^3\Pi_u^-$ and $j^3\Pi_g^- - c^3\Pi_u^-$ band systems are amongst those that Astashkevich et al. [18] have recently proposed as being candidates for use as molecular plasma diagnostics. Citing a lack of high-quality ab initio calculations for the rovibronic transition probabilities they rely on semiempirical methods to obtain these quantities. Using their semiempirical results they obtain a significant degree of success in obtaining reasonable plasma temperatures, which is *not* the case when they use Hönl–London factors. Therefore, there still seems to be a need for more theoretical work in this area, in particular given that they note some discrepancy between the plasma temperatures extracted from the $i^3\Pi_g^-$ and $I^1\Pi_g^-$ systems as compared with the other systems they consider.

4. Rovibronic calculations

The rovibronic calculations were performed in the same manner as described in ref. 2, suitably modified to account for our use of the $\tilde{\mu}$-defect. For the two sets of calculation we report here, the extent of the adiabatic vibrational basis sets of the molecular ion were

$$\begin{array}{lll} {}^3\Pi_u^- & (1\sigma_g)p: & \upsilon^+ = 0 \text{ to } 39 \\ {}^3\Pi_g^- \text{ with } {}^3\Delta_g^- & (1\sigma_g)d: & \upsilon^+ = 0 \text{ to } 44 \\ & (1\sigma_u)p: & \upsilon^+ = 0 \text{ to } 109 \end{array}$$

The adiabatic vibrational wave functions of H_2^+ and its isotopomers were calculated by the Numerov–Cooley technique [19] on a grid of 999 points covering the range $R = 0.1$ to 12.0 a.u., using the $(1\sigma_g)$ potential curve from Wind [20] and the $(1\sigma_u)$ potential curve from Bates et al. [21], supplemented with several values from Madsen and Peek [22] to extend the range to 12.0 a.u. The adiabatic corrections used for the $(1\sigma_g)$ state of H_2^+ were obtained by adding columns $H_1 + H_2$ of Bishop and Wetmore's Table 1 [23], while those used for the $(1\sigma_u)$ state are the ones calculated as described in Appendix B of ref. 2. To obtain adiabatic corrections for HD^+ and D_2^+ those of H_2^+ were adjusted for the different reduced masses.

To account for nonadiabatic, radiative, and relativistic effects on the vibrational levels of the $(1\sigma_g)$ state we have used vibrational energies from ab initio work on H_2^+ and its isotopomers, whenever these energy levels lay within 3.0 cm^{-1} of the Numerov–Cooley calculated values. (The ionic vibrational basis functions used in the MQDT calculation are restricted to $R \leq 12.0$ a.u. and therefore for higher υ values rise to higher energy than the strictly physical vibrational levels). We used $(1\sigma_g)$ state vibrational energy levels for H_2^+ from Moss [24], HD^+ levels from Moss [25], and D_2^+ levels from Wolniewicz and Orlikowski [26].

The "normal mass effect" was handled in the manner described in Sect. III.C.2 of ref. 2, while the "specific mass effect," described in Sect. III.C.3 of ref. 2, would require knowledge of that portion of the adiabatic corrections that Wolniewicz and Dressler [27] refer to as H_3', and its decomposition to particular channels. This information is not available for the triplet manifold and, therefore, we neglect it. We return to this point below.

The MQDT calculations yield absolute rovibronic energies. We convert these to energies relative to the $\upsilon = 0$, $N = 0$ level of the <1A> X ground state using the ground-state dissociation energies of Wolniewicz [28] in combination with the ionization energies of hydrogen and deuterium from Erickson [29]. We have adjusted Erickson's ionization energies for improvements in the value of the electron mass, the hydrogen and deuterium masses, and the Rydberg constants, values for which we have taken from CODATA 86 [30].

To handle the energy dependence of the quantum defects the appropriate intermediate rovibronic matrices were first calculated on a grid of three energies, $\epsilon = -0.250$, -0.125, and 0.0 a.u. and appropriate interpolations were performed during the course of the calculation. Note that the method of handling the energy dependence of the quantum defects in the rovibronic treatment must be carefully matched to that of the electronic treatment which was used to extract the quantum defects in the first place!

We have attempted to gather all currently available experimental rovibronic levels for the $^3\Pi_u^-$, $^3\Pi_g^-$, and $^3\Delta_g^-$ symmetries of H_2, HD, and D_2. Although some spectroscopic data exist for these symmetries for tritiated isotopomers, apparently this has not yet resulted in term values. The comparison of the MQDT calculations with the experiment data is presented in Tables 1 and 2 (for $^3\Pi_u^-$) and Table 3 (for $^3\Pi_g^-$ and $^3\Delta_g^-$). We now discuss the results for each symmetry in turn.

4.1. $^3\Pi_u^-$ states

In Table 1, we present all those H_2, HD, and D_2 $^3\Pi_u^-$ term values of which we are aware, and a comparison of the observed values with those we have calculated using MQDT. In *italics* we give MQDT calculated results for some terms whose values are not known experimentally. Note that highly accurate estimates for these unknown states could be made by taking into account the trends of the observed minus calculated (Obs. – Calc.) values for the known states.

For H_2 the bulk of the data is from Dieke's work as compiled by Crosswhite [34] (As Crosswhite's reference is known throughout the community as "Dieke's tables" for the remainder of this paper ref. 34 will be given as Dieke [34].) and includes six electronic states, $n = 2$ to 7. Dabrowski and Herzberg [35] have examined the <2c>c – <2a>a transitions. Combining their transition frequencies with the term values of the levels of the <2a>a state from Dieke [34] (lowered by 149.6 cm^{-1} as per ref. 36), we obtained terms for levels of the <2c>$^-$c state (averaging whenever more than one transition connected with a particular <2c>$^-$c level, or when the fine structure was partially resolved). These new values represent changes to a couple of Dieke's terms, with the new values in better agreement with the trend of the MQDT calculations. In addition four new <2c>$^-$c level term values result from the Dabrowski and Herzberg [35] transitions. For Table 1, we have chosen the terms derived from Dabrowski and Herzberg over those of Dieke, and, where available, those levels seen by Jungen et al. [37] supercede both these other sources.

Because of the importance of the high vibrational levels of the <2c>$^-$c metastable state as initial states for experimental probing of the triplet gerade manifold, we extend the range of "calculated-only" levels to rather high υ. Those levels given in ***bold italics*** in Table 1 correspond to initial states for the experiments considered in a companion article [5]. Although the term values for these high vibrational levels are not known, energy differences between some of them are known from the work of Lembo et al. [9]. Table 2 presents a comparison of the experimental and MQDT energy differences. The calculated values agree with experiment to within about one and a half times the experimental error.

Note that the quality of the MQDT values in Dieke's tables [34] made a typographical error for the <3c>$^-$d, $\upsilon = 4$, $N = 2$ level immediately apparent! Several other experimentally determined term values appear inconsistent with the current calculations. These have been underlined in the tables.

It is possible to "trick" the bound-state MQDT procedure into calculating levels lying above the ionization limit. This is done by the simple expedient of neglecting the open channels in the calculation. This procedure is worthwhile because it allows us to use the simple and extremely efficient bound-state

Table 1. Comparison of observed[a] and calculated term values for $^3\Pi_u^-$ states of H_2 (cm^{-1}).

H_2 $^3\Pi_u^-$ <2c>$^-$ c[b]

υ	*N* = 1	O–C[c]	*N* = 2	O–C	*N* = 3	O–C	*N* = 4	O–C	*N* = 5	O–C
0	94 941.58	–4.0[d]	95 062.33	–4.0[d]	95 242.37	–4.0[d]	95 480.26	–4.0[d]	95 774.38	–3.9[d]
1	97 280.60	–3.6[e]	97 395.55	–3.7[e]	97 567.03	–3.7[e]	97 793.62	–3.7[e]	98 073.75	–3.6[e]
2	99 497.27	–3.5[e]	99 606.78	–3.4[e]	99 769.97	–3.4[e]	99 985.52	–3.5[e]	100 251.95	–3.5
3	101 595.29	–3.3[e]	101 699.35	–3.3[e]	101 854.47	–3.2[e]	102 059.32	–3.3	102 312.55	–3.2
4	103 577.28	–2.9[f]	103 677.21	–1.8[f]	103 822.83	–3.3[g]	*104 020.50*		*104 260.63*	
5	105 445.50	–2.3[g]	105 539.99	–1.3[g]	105 677.87	–2.8[g]	*105 864.67*		*106 091.91*	
6	*107 202.23*		*107 290.55*		*107 422.09*		*107 595.78*		*107 810.22*	
7	***108 843.78***		*108 926.88*		*109 050.64*		*109 214.00*		*109 415.61*	
8	***110 371.84***		*110 449.68*		*110 565.56*		*110 718.47*		*110 907.06*	
9	***111 784.61***		*111 857.08*		*111 964.91*		*112 107.12*		*112 282.40*	
10	***113 079.11***		*113 146.01*		*113 245.52*		*113 376.65*		***113 538.10***	
11	***114 250.67***		*114 311.73*		***114 402.49***		*114 521.96*		*114 668.85*	
12	***115 292.74***		*115 347.60*		*115 429.00*		*115 535.99*		*115 667.25*	
13	*116 196.48*		*116 244.58*		*116 315.82*		*116 409.20*		*116 523.38*	
14	*116 950.56*		*116 991.21*		*117 051.19*		*117 129.49*		*117 224.69*	

υ	*N* = 6	O–C	*N* = 7	O–C	*N* = 8	O–C	*N* = 9	*N* = 10
0	96 122.49	–4.0	96 522.62	–3.7	96 956.33	–19.0	*97 470.45*	*98 008.81*
1	98 405.16	–3.7	98 785.80	–3.8	*99 216.88*		*99 688.08*	*100 200.29*
2	100 567.25	–3.5	100 929.21	–3.6	*101 339.11*		*101 787.03*	*102 273.78*
3	*102 615.30*		*102 959.13*		*103 344.88*		*103 770.01*	*104 231.84*
4	*104 544.71*		*104 870.67*		*105 236.25*		*105 638.99*	*106 076.29*

H_2 $^3\Pi_u^-$ <3c>$^-$ d

υ	*N* = 1	O–C	*N* = 2	O–C	*N* = 3	O–C	*N* = 4	O–C	*N* = 5	O–C
0	111 754.49	–0.8	111 872.26	–0.8	112 047.76	–0.8	112 279.62	–0.8	112 566.06	–0.9
1	113 994.73	–0.8	114 106.49	–0.8	114 273.07	–0.7	114 493.10	–0.7	114 764.97	–0.7
2	116 109.55	–0.7	116 215.56	–0.7	116 373.48	–0.7	116 582.05	–0.7	116 839.78	–0.7
3	118 103.04	–0.7	118 203.43	–0.7	118 352.96	–0.6	118 550.41	–0.7	118 794.26	–0.8
4	119 978.54	–0.6	120 073.44	–0.5[h]	120 214.79	–0.5	120 401.13	–0.8	*120 632.43*	
5	121 738.66	–0.4	121 828.18	–0.4	121 961.19	–0.7	122 137.59	–0.3	*122 355.16*	
6	123 385.38	–0.2	123 469.58	–0.2	123 594.59	–0.6	123 759.91	–0.7	*123 964.90*	
7	124 919.89	0.0 [υ1]	124 998.79	0.0 [υ1]	125115.94	–0.4 [υ1]	*125 271.38*	[υ1]	*125 462.66*	[υ1]
8	126 342.49	0.1 [υ2]	*126 416.00*	[υ2]	*126 525.62*	[υ2]	*126 670.21*	[υ2]	*126 848.47*	[υ2]
9	127 652.55	0.3 [υ2]	*127 720.57*	[υ2]	*127 822.24*	[υ2]	*127 956.16*	[υ2]	*128 121.21*	[υ2]

υ	*N* = 6	O–C	*N* = 7	O–C	*N* = 8	O–C
0	112 905.21	–0.8	*113 295.21*		113 736.53	4.6
1	115 086.74	–0.6	115 455.83	–0.7	115 869.88	–0.8
2	117 144.62	–0.7	*117 495.12*		*117 887.43*	

Table 1. (*continued*).

H_2 $^3\Pi_u^-$ <4c>$^-$ k

υ	*N* = 1	O–C	*N* = 2	O–C	*N* = 3	O–C	*N* = 4	O–C	*N* = 5	O–C
0	117 406.30	–2.8	117 523.08	–2.8	117 697.12	–2.8	117 927.05	–2.8	118 211.09	–2.8
1	119 615.94	–3.0	119 726.68	–3.0	119 891.69	–3.0	120 109.63	–3.1	120 378.96	–3.0
2	121 699.81	–3.3	121 804.70	–3.3	121 960.92	–3.4	122 167.36	–3.4	122 422.34	–3.4
3	123 661.90	–3.6	123 761.15	–3.6	123 908.98	–3.7	124 104.24	–3.7	124 345.76	–3.5
4	125 504.94	–4.0 [υ1]	125 598.62	–4.0	125 738.24	–3.9 [υ1]	125 922.46	–4.0 [υ1]	126 149.92	–4.1 [υ1]
5	127 232.64	–4.3 [υ2]	127 320.83	–4.4	127 452.40	–4.3 [υ2]	*127 630.25*	[υ2]	*127 844.47*	[υ2]
6	128 846.48	–4.8 [υ3]	128 929.03	–5.2	129 052.94	–4.8 [υ3]	*129 220.70*	[υ3]	*129 421.81*	[υ3]
7	130 347.31	–6.2 [υ4]	*130 430.95*	[υ4]	*130 546.73*	[υ4]	*130 699.25*	[υ4]	*130 887.38*	[υ4]

υ	*N* = 6	O–C	*N* = 7	O–C
0	118 537.19	–12.9	118 933.31	–2.6

H_2 $^3\Pi_u^-$ <5c>$^-$ n Note: calculations denoted υ1 are less precise since neglected $\upsilon^+ = 0$ level contributes significantly here.

υ	*N* = 1	O–C	*N* = 2	O–C	*N* = 3	O–C	*N* = 4	O–C	*N* = 5	O–C
0	119 982.34	–2.2	120 098.89	–2.0	120 272.20	–2.2	120 501.29	–2.2	120 784.35	–2.3
1	122 178.04	–1.9	122 289.31	–2.0	122 454.57	–2.1	122 672.38	–2.3	122 941.22	–2.4
2	124 248.03	–10.0 [j]	124 352.42	–10.1 [υ1,j]	124 506.56	–11.7 [υ1,j]	124 709.75	–14.2 [υ1,j]	124 959.92	–18.0 [υ1,j]
3	126 197.10	–13.0 [υ1]	126 294.51	–14.4 [υ1]	126 438.55	–17.5 [υ1]	126 629.58	–20.8 [υ1]	*126 890.39*	[υ1]

H_2 $^3\Pi_u^-$ <6c>$^-$ u

υ	*N* = 1	O–C	*N* = 2	O–C	*N* = 3	O–C	*N* = 4	O–C	*N* = 5	O–C
0	121 367.33	–1.4	121 483.66	–1.3	121 656.77	–1.4	121 884.81	–2.2	122 166.35	–3.4
1	123 556.04	–10.1 [k]	*123 670.35*		*123 836.28*		*124 053.67*		*124 321.50*	

H_2 $^3\Pi_u^-$ <7c>$^-$

υ	*N* = 1	O–C	*N* = 2	O–C	*N* = 3	O–C
0	122 203.80	–1.6	*122 320.48*		122 492.75	0.2

HD $^3\Pi_u^-$ <2c>$^-$ c

υ	*N* = 1	O–C	*N* = 2	O–C	*N* = 3	O–C	*N* = 4	O–C	*N* = 5	O–C
0	95 066.63	–2.9	95 157.65	–2.9	95 293.52	–2.9	95 473.48	–2.9	95 696.55	–2.9
1	97 107.50	–2.8	97 194.78	–2.8	97 325.07	–2.8	97 497.62	–2.8	97 711.48	–2.8
2	99 056.17	–2.7	99 139.80	–2.7	99 264.64	–2.7	99 429.96	–2.7	99 634.84	–2.7
3	100 914.99	–2.6	100 995.07	–2.6	101 114.60	–2.6	101 272.87	–2.6	101 469.00	–2.6
4	*102 688.30*		*102 764.88*		*102 879.19*		*103 030.54*		*103 218.07*	

υ	*N* = 6	O–C	*N* = 7	O–C	*N* = 8	O–C	*N* = 9	*N* = 10
0	95 961.53	–2.9	96 267.06	–2.8	96 611.65	–2.6	*96 995.88*	*97 412.87*
1	97 965.49	–2.7	98 258.35	–2.7	98 588.58	–2.6	*98 957.29*	*99 356.36*
2	99 878.16	–2.7	100 158.64	–2.7	*100 477.44*		*100 827.61*	*101 210.12*
3	101 701.89	–2.6	101 970.32	–2.6	*102 275.34*		*102 610.31*	*102 976.15*
4	*103 440.71*		*103 697.25*		*103 986.30*		*104 306.36*	*104 655.83*

Table 1. (*continued*).

D_2 $^3\Pi_u^-\langle 2c\rangle^-$ c

υ	*N* = 1	O–C	*N* = 2	O–C	*N* = 3	O–C	*N* = 4	O–C	*N* = 5	O–C
0	95 215.52	–2.0[l]	95 276.54	–2.0[l]	95 367.79	–2.0[l]	95 488.88	–2.0[l]	95 639.45	–2.0[l]
1	96 897.10	–1.8[m]	96 956.09	–1.8[m]	97 044.15	–1.9[m]	97 161.28	–1.8[m]	97 306.50	–2.0[m]
2	98 516.54	–1.7[m]	98 573.50	–1.7[m]	98 658.68	–1.7[m]	98 771.64	–1.8[m]	98 912.23	–1.7
3	100 075.35	–1.6[m]	100 130.25	–1.7[m]	100 212.48	–1.7[m]	100 321.62	–1.7[m]	100 457.41	–1.6
4	101 574.69	–1.5[m]	101 627.68	–1.6[m]	101 707.15	–1.5[m]	101 812.35	–1.6[m]	101 943.26	–1.5
5	103 015.45	–1.6[m]	103 066.63	–1.5[m]	103 143.15	–1.5[m]	103 244.67	–1.5[m]	*103 372.36*	
6	104 398.65	–1.5[g]	104 447.97	–1.5[g]	104 521.64	–1.5[g]	104 619.46	–1.5[g]	*104 742.46*	
7	*105 726.25*		*105 773.69*		*105 844.60*		*105 938.70*		*106 055.62*	

υ	*N* = 6	O–C	*N* = 7	O–C	*N* = 8	O–C	*N* = 9	O–C	*N* = 10	O–C
0	95 818.85	–2.0[l]	96 026.49	–2.0[l]	96 261.54	–2.0[l]	96 523.43	–3.2	96 810.90	–3.3
1	97 480.05	–1.8	97 680.61	–1.8	97 907.77	–1.8	98 160.72	–3.1	98 438.51	–3.1
2	99 079.72	–1.6	99 273.43	–1.6	99 492.69	–1.7	99 737.00	–3.0	100 004.94	–3.2
3	100 618.98	–1.6	100 806.05	–1.6	101 017.71	–1.6	101 253.55	–2.9	101 512.82	–2.4
4	*102 100.72*		*102 281.12*		*102 485.33*		*102 714.06*		*102 963.64*	

υ	*N* = 11	O–C	*N* = 12	O–C	*N* = 13	O–C	*N* = 14	O–C	*N* = 15	O–C
0	97 123.47	–3.1	97 459.43	–3.4	97 818.66	–3.1	98 199.37	–3.1	98 600.75	–3.1
1	98 739.79	–3.5	*99 068.04*		*99 414.74*		*99 782.37*		*100 169.85*	
2	100 257.58	–41.9	*100 612.93*		*100 947.58*		*101 302.38*		*101 676.30*	
3	*101 796.32*		*102 098.75*		*102 421.56*		*102 763.76*		*103 124.34*	

D_2 $^3\Pi_u^-\langle 3c\rangle^-$ d

υ	*N* = 1	O–C	*N* = 2	O–C	*N* = 3	O–C	*N* = 4	O–C	*N* = 5	O–C
0	112 043.96	–0.5	112 103.52	–0.4	112 192.50	–0.5	112 310.66	–0.4	112 457.46	–0.4
1	113 656.49	–0.5	113 713.89	–0.4	113 799.69	–0.4	113 913.57	–0.4	114 055.07	–0.4
2	115 205.29	–0.4	115 260.56	–0.4	115 343.23	–0.4	115 452.95	–0.4	115 589.28	–0.4
3	116 691.87	–0.4	116 745.12	–0.4	116 824.69	–0.5	116 930.35	–0.4	117 061.63	–0.4
4	118 117.65	–0.4	118 168.90	–0.4	118 245.51	–0.4	118 347.17	–0.4	118 473.48	–0.4
5	119 483.88	–0.4	119 533.16	–0.4	119 606.83	–0.4	119 704.57	–0.4	119 826.00	–0.4
6	120 791.58	–0.4	120 838.92	–0.4	120 909.70	–0.4	121 003.61	–0.4	121 120.25	–0.4
7	122 041.66	–0.3	122 087.09	–0.3	122 155.02	–0.3	122 245.10	–0.3	122 357.02	–0.3
8	123 234.81	–0.2	123 278.38	–0.2	123 343.49	–0.2	123 429.79	–0.2	123 537.00	–0.3
9	124 371.51	–0.2	124 413.20	–0.1	124 475.48	–0.1	124 557.93	–0.3	124 661.48	0.6
10	125 452.15	–0.1 [υ1]	125 492.00	–0.1 [υ1]	125 551.54	0.0 [υ1]	125 630.35	–0.1 [υ1]	125 728.39	0.0 [υ1]

υ	*N* = 6	O–C	*N* = 7	O–C	*N* = 8	O–C	*N* = 9	O–C	*N* = 10	O–C
0	112 632.41	–0.4	112 834.78	–0.4	113 063.92	–0.4	113 318.94	–1.7	113 598.97	–1.7
1	114 223.70	–0.4	114 418.72	–0.4	114 639.53	–0.4	114 885.25	–1.7	115 155.09	–1.7
2	115 751.71	–0.4	115 939.57	–0.4	116 152.26	–0.4	116 388.89	–1.8	116 648.75	–1.8
3	117 218.08	–0.4	117 398.88	–0.4	117 603.61	–0.4	117 831.37	–1.9	118 081.48	–1.8
4	118 623.97	–0.4	118 797.96	–0.4	118 994.92	–0.4	119 213.47	–2.4	119 454.55	–1.8
5	119 970.67	–0.4	120 137.96	–0.4	120 327.27	–0.4	120 537.14	–2.6	120 768.58	–2.2
6	121 259.19	–0.4	121 419.90	–0.3	121 602.07	0.1	*121 805.70*		*122 027.54*	
7	122 490.65	0.0	122 644.46	–0.3	122 818.97	–0.1	*123 014.53*		*123 227.22*	
8	123 664.74	–0.2	*123 812.59*		*123 979.56*		*124 166.80*		*124 370.41*	

Table 1. (*continued*).

υ	*N* = 11	O–C	*N* = 12	O–C	*N* = 13	O–C	*N* = 14	O–C	*N* = 15	O–C
0	*113 904.70*		114 230.17	–1.7	114 579.33	–1.6	114 949.36	–1.6	115 339.42	–1.4
1	115 448.02	–1.7	*115 764.83*		116 099.44	–1.7	116 455.79	–1.7	*116 832.92*	
2	*116 932.60*		117 230.52	–5.5	*117 559.74*		*117 902.79*		*118 264.11*	
3	118 352.84	–2.0	118 641.80	–4.9	118 951.67	–6.5	119 278.92	–9.2	*119 635.66*	
4	119 715.47	–2.0	119 993.49	–4.6	*120 297.54*		*120 614.71*		*120 948.63*	

D_2 $^3\Pi_u^-\langle 4c\rangle^-$ k

υ	*N* = 1	O–C	*N* = 2	O–C	*N* = 3	O–C	*N* = 4	O–C	*N* = 5	O–C
0	117 700.76	–2.6	117 759.79	–2.6	117 848.08	–2.6	117 965.24	–2.6	118 110.87	–2.6
1	119 292.45	–2.8	119 349.34	–2.8	119 434.38	–2.8	119 547.21	–2.8	119 687.45	–2.8
2	120 819.98	–3.0	120 874.72	–3.0	120 956.58	–3.1	121 065.30	–3.0	121 200.18	–3.1
3	122 284.80	–3.3	122 337.50	–3.3	122 416.24	–3.3	122 520.82	–3.3	122 650.67	–3.3
4	123 688.47	–3.5	123 739.12	–3.6	123 814.83	–3.6	123 915.43	–3.5	124 040.22	–3.6
5	125 032.31	–3.8 $^{\upsilon1}$	125 080.91	–3.8 $^{\upsilon1}$	125 153.59	–3.9 $^{\upsilon1}$	125 250.72	–3.4 $^{\upsilon1}$	125 370.29	–3.8 $^{\upsilon1}$
6	126 317.43	–4.1 $^{\upsilon2}$	126 364.16	–4.1 $^{\upsilon2}$	*126 438.20*	$^{\upsilon2}$	126 526.85	–4.1 $^{\upsilon2}$	*126 646.23*	$^{\upsilon2}$

υ	*N* = 6	O–C	*N* = 7	O–C	*N* = 8	O–C	*N* = 9	O–C	*N* = 10	O–C
0	118 284.31	–2.6	118 485.05	–2.6	118 712.22	–2.6	118 964.91	–4.0	119 241.48	–5.1
1	119 854.54	–2.8	120 047.73	–2.9	120 266.51	–2.9	*120 514.19*		*120 781.48*	
2	121 361.08	–3.0	121 566.72	16.6	121 776.67	16.0	*121 996.31*		*122 253.56*	
3	122 805.42	–3.3	122 994.12	6.4	*123 190.24*		*123 416.98*		*123 664.33*	
4	124 189.00	–3.6	124 360.62	–4.0	*124 559.32*		*124 777.35*		*125 015.04*	
5	125 513.06	–3.9 $^{\upsilon1}$	*125 682.26*	$^{\upsilon1}$	125 869.24	$^{\upsilon1}$	*126 078.72*	$^{\upsilon1}$	126 306.97	$^{\upsilon1}$

D_2 $^3\Pi_u^-\langle 5c\rangle^-$ n

υ	*N* = 1	O–C	*N* = 2	O–C	*N* = 3	O–C	*N* = 4	O–C	*N* = 5	O–C
0	120 277.83	–2.0	120 336.67	–2.1	120 424.67	–2.1	120 541.50	–2.0	120 686.65	–2.0
1	121 862.81	–2.3	121 919.47	–2.3	122 004.20	–2.3	122 116.68	–2.3	122 256.32	–2.5
2	123 383.90	–2.6	123 438.51	–2.6	123 520.09	–2.6	123 628.40	–2.6	123 762.90	–2.7
3	124 843.82	–3.0 $^{\upsilon1}$	124 896.47	–3.0 $^{\upsilon1}$	124 975.11	–3.0 $^{\upsilon1}$	125 079.44	–3.1 $^{\upsilon1}$		

υ	*N* = 6	O–C	*N* = 7	O–C	*N* = 8	O–C
0	120 859.52	–2.1	121 059.52	–2.2	121 286.11	–2.0
1	*122 425.34*		*122 617.98*		*122 836.03*	
2	123 923.35	–2.6	*124 111.46*		*124 321.42*	

D_2 $^3\Pi_u^-\langle 6c\rangle^-$ u

υ	*N* = 1	O–C	*N* = 2	O–C	*N* = 3	O–C	*N* = 4	O–C	*N* = 5	O–C
0	121 664.40	–1.4	121723.20	–1.4	121811.15	–1.3	121927.66	–1.4	122 072.52	–1.4
1	123 243.96	–1.5	123300.56	–1.4	123385.17	–1.4	123497.34	–1.5	123 636.74	–1.5
2	124 757.58	–1.4 $^{\upsilon1}$	124811.91	–1.4 $^{\upsilon1}$	124893.19	–1.4 $^{\upsilon1}$	125001.07	–1.3 $^{\upsilon1}$	125 134.79	–1.5 $^{\upsilon1}$

υ	*N* = 6	O–C
0	*122 246.60*	
1	123 802.91	–1.5
2	125 294.25	–1.5 $^{\upsilon1}$

Table 1. (*concluded*).

D_2 $^3\Pi_u^-$ <7c>$^-$

υ	$N = 1$	O–C	$N = 2$	O–C	$N = 3$	O–C	$N = 4$	O–C	$N = 5$	O–C
0	122 497.50	–0.4	122555.58	–1.0	*122 644.42*		122759.95	–1.0	122 904.27	–1.5
1	124 075.80	–1.1	124132.44	–1.1	124 216.91	–1.0	124329.15	–1.1	124 468.50	–1.2

D_2 $^3\Pi_u^-$ <8c>$^-$

υ	$N = 1$	O–C	$N = 2$	O–C	$N = 3$	O–C	$N = 4$	O–C
0	123 034.38	–0.7	123093.14	–0.7	123 180.85	–0.8	123297.22	–0.9

D_2 $^3\Pi_u^-$ <9c>$^-$

υ	$N = 1$	O–C	$N = 2$	O–C	$N = 3$	O–C	$N = 4$	O–C	$N = 5$	O–C
0	123 402.28	–0.5	123460.69	–0.8	123 548.62	–0.6	123664.94	–0.7	123 809.06	–1.3

[a] Observed values in normal type. Calculated values (when no observed is available) are given in *italics* with no O–C values. Values in ***bold italics*** are those used in a companion article [5]. Underlined observed values may be questionable (as indicated by current results). The observed values are taken from various references as follows:

H_2: All data are from Dieke [34] except those flagged with a "*d*", which are from Jungen et al. [37], or those superscripted with an "*e*, *f*, or *g*", which are derived from Dabrowski and Herzberg [35]. These latter ones were derived by adding the transitions reported in their Tables I and III to the <2a> a-state terms of Dieke, averaging [34] where more than one value is available (partially resolved fine structure values are included in the average). Note that all terms from Dieke [34] have been lowered by 149.6 cm^{-1} [36].

HD: All data are from Table II of Keiding and Bjerre [38], combined with the energy, 97 107.50 cm^{-1}, of the <2c>$^-$ c, υ=0, N=1 level from Table VI of Kim and Mazur [39] (as interpreted from their eq. (6), note that this value appears to be misprinted in their abstract and conclusions).

D_2: All data are from Freund et al. [12], except those superscripted with an "*l*" (which are from Jungen et al. [40]) or an "*m*" (which are derived from Dabrowski and Herzberg [35] by adding the transitions reported in their Table II to the <2a> a state terms of Freund et al., averaging where more than one value is available.

[b] States are labelled with Dieke's notation given in triangular brackets, e.g., <2c>, where "*n*c" corresponds to "$np\pi_u$". The Kronig parity is added as a superscript, e.g., <2c>$^-$. The traditional or Herzberg notation, where such exists, is given by the following characters, e.g., <2c>$^-$ c.

[c] Observed – calculated values in cm^{-1}.

[d] Terms from Table I of Jungen et al. [37].

[e] Terms derived from Dabrowski and Herzberg [35] in agreement with those of Dieke [34].

[f] These values, derived from Dabrowski and Herzberg [35] differ significantly from the values of Dieke [34].

[g] These new values, derived from Dabrowski and Herzberg [35] represent terms not reported in Dieke [34] (H_2) or Freund et al. [12] (D_2).

[h] There is a typographical error in Dieke [34]. "120 233.04" there should read "120 223.04", which when corrected [36] by –149.6 cm^{-1} becomes 120 073.44. This value then agrees with the transitions tabulated in Dieke.

[i] A calculated n^* = 22, υ=0, N=1 level is at 124 247.9. If we were to take this as the assignment we would then have O–C = 0.1.

[j] These levels appear in the transition tables of Dieke [34], but not in the term value tables there.

[k] A calculated n^* = 11, υ=0, N=1 level lies at 123 557.0. If we were to take this as the assignment we would then have O–C = –1.0.

[l] Terms from Table I of Jungen et al. [40].

[m] D_2 terms derived from Dabrowski and Herzberg [35].

[υ] υ1, υ2, ... denote calculated levels lying near or above the lowest involved ionization limit. These were obtained with ionic vibrational basis sets starting with υ^+=1, 2, ..., respectively.

MQDT for states that are only weakly preionized, rather than having to use the heavier machinery of the full continuum versions of MQDT. In practice this means progressively deleting the channels built on the $\upsilon^+ = 0$, $\upsilon^+ = 0$ and 1, and $\upsilon^+ = 0$, 1, and 2, etc. levels of H_2^+, as we search for ever higher rovibronic levels. Term values calculated in this way are flagged in the tables by indicating the lowest H_2^+ vibrational level actually included. Thus, for example, "$\upsilon 1$" indicates that channels built on $\upsilon^+ = 0$ have been deleted from the calculation. In practice this trick has occasionally also proven useful in calculating levels lying in the congested region just below an ionization threshold.

This approach will succeed in so far as the open channels neglected in the calculation contribute little to the level being calculated. Thus, highly vibrationally excited levels of low-*n* states will be well represented while lowly excited vibrational levels of higher *n* states will be only poorly represented.

Table 2. Comparison of observed[a] and calculated term value differences for high vibrational levels of the $^3\Pi_u^-$ <2c>$^-$ c state of H_2 (cm^{-1}).

H_2 $^3\Pi_u^-$ <2c>$^-$								
υ	$N = 1$	O–C[b]	$N = 3$	O–C	$N = 5$	O–C	$N = 7$	O–C
7	–2943.0 ± 1.4	–2.2						
8	–1413.8 ± 1	–1.0						
9	0.0				496.9 ± 2	–0.9	940.4 ± 2	–0.6
10	1295.5 ± 1	1.0			1753.6 ± 2	0.1	2160.9 ± 2	0.0
11	2467.4 ± 1	1.3	2619.1 ± 1	1.2	2884.4 ± 2	0.2	3254.4 ± 2	0.6
12	3509.5 ± 1	1.4	3645.7 ± 1	1.3	3883.8 ± 2	1.2		

[a] Observed values are the energies relative to the <2c>$^-$ c, $\upsilon = 9$, $N = 1$ level. Values from Table I of Lembo et al. [9] and given with experimental uncertainty.
[b] Observed – Calculated values in cm^{-1}.

This can be seen, for example, in the Obs. – Calc. values for H_2. The preionized levels of the $<4c>^-$k state begin at $\upsilon = 4$. For these levels the neglect of the channels built on the lowest vibrational states of H_2^+ has little effect on the quality of the calculated energies as can be seen by examining the trend of the Obs. – Calc. values. For the $<5c>^-$n state, however, the preionized levels begin at $\upsilon = 2$ and the neglect of the lowest channels has a significant effect on the Obs. – Calc. values.

For the HD isotopomer, we obtained experimental term values for levels of the $<2c>^-$c state by combining the relative term values of Keiding and Bjerre [38] with the term value for the $<2c>^-$c, $\upsilon = 0$, $N = 1$ level of 97 107.50 cm^{-1} from Kim and Mazur [39].

For D_2, a plethora of term values are available that include levels of eight electronic states, $n = 2$ to 9. The bulk of this data is from the work of Dieke as compiled and corrected by Freund et al. [12]. As was the case with H_2, term values for some levels of the $<2c>^-$c state can be derived from the transitions of Dabrowski and Herzberg [35]. These include four newly identified levels. Also, in a work subsequent to theirs for H_2, Jungen et al. [40] obtained precise term values for several levels of the $<2c>^-$c state of D_2.

One notable feature of the $<2c>^-$c calculations are the relatively large Obs. – Calc. values for this lowest member of the Rydberg series. These errors, however, are mass dependent. For the $\upsilon = 0$, $N = 1$ level the Obs. – Calc. values are –4.0 cm^{-1} for H_2, –2.9 cm^{-1} for HD, and –1.8 cm^{-1} for D_2 This sequence extrapolates linearly to +0.3 cm^{-1} at infinite mass, corresponding to the limit of the accuracy of the present calculations. We, therefore, ascribe the Obs. – Calc. values to the neglected specific mass effect discussed at the top of this section. In the case of the singlet <2C>C state of H_2 the neglected H_3' term has the value 4.165 cm^{-1}, almost exactly corresponding to the MQDT Obs. – Calc. error in the triplet state, although of the opposite sign in agreement with the expected sign reversal of H_3' between triplet and singlet, at least in the asymptotic limit.[5]

4.2. $^3\Pi_g^-$ and $^3\Delta_g^-$ states

Table 3 is the gerade equivalent to Table 1, presenting a comparison of the MQDT calculations with experimental term values for all states for which we have been able to identify experimental results. We have made an effort to collect all known terms for these symmetries. As noted in Table 3, we have determined many of these term values in a consistent way from a variety of transitions reported in the literature. Many of the term values so derived represent significant changes to the term values of Dieke [34] and are flagged with a Δ.

[5] L. Wolniewicz. Private communication.

Table 3. Comparison of observed[a] and calculated term values for $^3\Pi_g^-$ and $^3\Delta_g^-$ states of H_2 (cm^{-1}).

H_2 $^3\Pi_g^-$ <3e>⁻ i[b]

υ	N = 1	O–C[c]	N = 2	O–C	N = 3	O–C	N = 4	O–C	N = 5	O–C
0	112 066.65	–0.7[d]	112 140.82	–0.8[d]	112 264.88	–1.0[d]	112 441.77	–1.1[d]	112 671.67	–1.4[e]
1	114 179.22	–0.2[e]	114 259.06	–0.2[e]	114 384.38	–0.4[e]	114 557.69	–0.6[e]	114 779.65	–0.7[e]
2	116 145.57	0.0[e]	116 227.95	0.1[e]	116 353.23	–0.2[e]	116 523.39	–0.1[e]	116 737.95	–0.5[e]
3	117 958.86	–0.1[e]	118 041.08	–0.2[e]	118 164.91	–0.4[e]	118 330.71	–0.5[e]	*118 539.02*	
4	119 596.4	–0.5[f]	119 675.7	–1.2[f]	119 797.4	0.5[f]	119 955.	–1.3[f]	120 153.	–2.[f]
5	120 955.	–13.[f]	121 042.	14.[f]	121 148.	–37.[f]				

υ	N = 6	O–C	N = 7	O–C	N = 8	O–C
0	112 953.92	–1.4	113 284.74	–3.5	*113 664.22*	
1	115 050.81	0.2	*115 367.83*		*115 730.08*	
2	116 997.41	–0.5	117 300.10	–0.8	117 644.71	–1.1
3	118 792.	4.[g]				
4	120 393.	2.[f]				

n = 2 dissociation limit

H_2 $^3\Pi_g^-$ <4e>⁻ r

υ	N = 1	O–C	N = 2	O–C	N = 3	O–C	N = 4	O–C	N = 5	O–C
0	117 587.66	–0.4[hΔ]	117 599.44	–0.3[h]	117 706.10	–0.4[h]	117 874.13	–0.4[h]	118 099.26	–0.5[h]
1	119 739.35	–0.3[hΔ]	119 770.24	0.0[h]	119 873.14	–0.1[h]	120 033.08	–0.2[h]	120 246.72	–0.3[hΔ]
2	121 756.54	–0.3[hΔ]	121 802.95	–0.1[hΔ]	121 904.43	–0.1[hΔ]	122 057.75	–0.1[hΔ]	**122 260.82**	**–0.2**[iΔ]
3	123 366.57	–271.0	**123 696.11**	**0.2**[iΔ]	**123 798.40**	**0.0**[iΔ]	**123 946.66**	**0.0**[iΔ]	**124 140.49**	**–0.2**[i]
8			*129 279.	–12.[υ3,j]						
9			*130 132.5	–11.4[υ3,j]			*130 321.0	—[j]		
10			*130 916.9	–14.9[υ4,j]			***131 127.6**	–10.4[υ4,j]		
11			*131 662.0	–11.1[υ4,j]						
12			*132 307.4	–1.3[υ3,j]						

υ	N = 6	O–C	N = 7	O–C
0	118 378.95	–0.6[h]	*118 711.31*	
1	120 511.79	–0.3[hΔ]	120 826.10	–0.1
2	122 512.02	–0.3[hΔ]	*122 809.73*	

Table 3. (*continued*).

H_2 $^3\Delta_g^-$ <3f>$^-$ j

υ		N = 2	O–C	N = 3	O–C	N = 4	O–C	N = 5	O–C
0		112 513.95	0.6[d]	112 732.18	1.0[d]	113 007.64	1.4	113 336.26	1.7
1		114 706.80	1.4[e]	114 904.61	1.9[e]	115 158.20	2.4[e]	115 463.46	2.9[e]
2	n = 2 dissociation	116 776.23	2.3[e]	116 954.76	2.7[e]	117 186.46	3.3[e]	117 467.14	3.6[e]
3	limit	*118 725.16	1.1[e]	118 885.31	1.6[e]	119 094.20	2.1[e]	119 348.28	2.5[e]
4		*120 555.6	2.2[f]	120 695.8	4.5[f]	120 863.4	–7.3[f]		
5		*122 279.	–3.[k]	122 430.	27.[g]	122 598.	–70.[k]		
6		*123 887.	–3.[k]	124 011.	0.[g]	124 172.	20.[g]		
7		*125 382.	1.[υ1,k]			125 651.	21.[υ1,g]		
8		*126 764.	31.[υ2,l]						
9		*128 034.	–11.[υ2,l]						
10		*129 188.	28.[υ3,l]						
11		*130 227.	4.[υ3,l]	130 309.	3.[υ3,l]				
12		*131 145.	13.[υ4,l]	131 215	25.[υ4,l]	*131 312.	47.[υ4,l]		
13		*131 935.	–32.[υ4,l]						
14		*132 588.	9.[υ5,l]						

υ	N = 6	O–C	N = 7	O–C
0	113 714.98	2.0	*114 138.32*	
1	115 816.67	3.3	116 214.64	3.8
2	117 793.36	4.0	118 161.58	4.3

H_2 $^3\Delta_g^-$ <4f>$^-$ s

υ	N = 2	O–C	N = 3	O–C	N = 4	O–C
0	117 830.35	0.1[h]	118 067.15	0.2[h]	118 352.97	0.3[h]
1	119 997.24	0.8[h]	120 226.08	0.3[hΔ]	120 491.70	0.7[hΔ]
2	122 042.65	1.2[hΔ]	122 246.82	1.2[hΔ]	122 493.10	1.1[hΔ]
3	123 969.04	1.0[hΔ]	124 155.12	1.1[hΔ]	124 387.56	1.3[hΔ]

HD $^3\Pi_g^-$ <3e>$^-$ i

υ	N = 1	O–C	N = 2	O–C	N = 3	O–C	N = 4	O–C	N = 5	O–C
0	112 178.84	–0.5	112 243.26	–0.6	112 344.04	–0.7	112 483.10	–0.9	112 661.32	–1.0
1	114 027.69	–0.2	114 093.72	–0.2	114 194.55	–0.4	114 331.52	–0.4	114 505.06	–0.5
2	115 767.54	0.2	115 833.87	0.3	115 933.87	0.1	116 068.28	0.0	116 237.26	–0.1
3	117 395.20	0.2	117 460.67	0.1	117 559.06	0.0	117 690.39	–0.1	117 854.58	–0.3

υ	N = 6	O–C	N = 7	O–C
0	112 878.63	–1.1	113 134.44	–1.2
1	114 715.20	–0.6	114 961.73	–0.4
2	116 440.64	–0.2	*116 678.26*	
3	*118 051.74*		118 280.00	–0.7

Table 3. (*continued*).

HD $^3\Pi_g^-$ <3f>$^-$ j

υ	*N* = 2	O–C	*N* = 3	O–C	*N* = 4	O–C	*N* = 5	O–C
0	112 616.21	0.4	112 774.26	0.7	112 977.88	1.0	113 224.27	1.2
1	114 536.56	0.8	114 682.05	1.2	114 871.25	1.6	115 101.74	2.0
2	116 362.68	2.1	116 496.52	2.4	116 671.70	2.7	116 886.19	3.0
3	118 096.83	1.4	118 219.68	1.6	118 380.95	1.8	118 579.06	2.2

D_2 $^3\Pi_g^-$ <3e>$^-$ i

υ	*N* = 1	O–C	*N* = 2	O–C	*N* = 3	O–C	*N* = 4	O–C	*N* = 5	O–C
0	112 317.02	0.6^m	112 365.63	0.6^m	112 439.17	0.4^m	112 538.36	0.3^m	112 663.32	0.2^m
1	113 845.27	1.7	113 893.63	1.7	113 966.40	1.5	114 063.94	1.4	114 186.37	1.3
2	115 301.01	1.5	115 348.78	1.5	115 420.39	1.5	115 516.05	1.4	115 635.72	1.4
3	116 683.29	1.2	116 730.18	1.2	116 800.31	1.1	116 893.90	1.1	117 010.64	1.0
4	<u>117 975.17</u>	<u>–13.2</u>	118 038.20	4.0	118 103.67	0.9	118 194.84	0.8	*118 307.76*	

υ	*N* = 6	O–C	*N* = 7	O–C	*N* = 8	O–C	*N* = 9	O–C	*N* = 10	O–C
0^m	112 814.42	0.1^m	112 991.34	–0.1^m	113 193.94	–0.5^m	113 421.56	–0.7	113 673.88	–1.7
1	114 333.69	1.2	114 505.73	1.1	114 702.24	0.7	114 922.61	0.6	115 166.51	–0.5
2	115 779.32	1.3	115 946.70	1.4	116 137.33	1.0	116 350.58	0.7	*116 586.77*	
3	117 150.42	1.0	117 313.01	0.9	117 496.80	–0.8	117 700.04	–4.6	*117 934.02*	

υ	*N* = 11	O–C	*N* = 12	O–C	*N* = 13	O–C	*N* = 14	O–C
0	113 949.56	–2.1	114 248.56	–2.0	114 568.94	–2.5	114 910.84	–2.4
1	115 433.06	–0.5	*115 722.09*		*116 031.59*		*116 361.19*	

D_2 $^3\Pi_g^-$ <4e>$^-$ r

υ	*N* = 1	O–C	*N* = 2	O–C	*N* = 3	O–C	*N* = 4	O–C	*N* = 5	O–C
0	117 837.77	5.3	117 866.15	1.4	117 924.81	0.1	118 013.00	0.0	118 129.64	0.0

υ	*N* = 6	O–C	*N* = 7	O–C
0	118 274.22	0.0	118 446.27	–0.1

D_2 $^3\Pi_g^-$ <3f>$^-$ j

υ	*N* = 2	O–C	*N* = 3	O–C	*N* = 4	O–C	*N* = 5	O–C
0	112 749.45	0.1^m	112 849.93	0.3^m	112 981.74	0.4^m	113 143.77	0.5^m
1	114 336.37	0.3	114 430.86	0.6	114 555.31	0.8	114 708.62	1.0
2	115 859.36	1.7	115 948.15	1.9	116 065.43	2.1	116 210.32	2.3
3	117 319.74	2.0	117 403.21	2.1	117 513.51	2.3	117 649.88	2.2
4	118 718.86	1.0	118 796.88	1.1	118 900.06	1.3	119 027.59	1.4
5	120 057.46	0.6						

Table 3. (*concluded*).

D_2 $^3\Pi_g^-$ <3f>$^-$ j

υ	*N* = 6	O–C	*N* = 7	O–C	*N* = 8	O–C	*N* = 9	O–C	*N* = 10	O–C
0	113 335.03	0.9[m]	113 553.58	1.0[m]	113 798.96	0.3	114 069.44	0.3	114 364.31	0.4
1	114 889.88	1.4	115 097.80	1.6	115 331.23	0.9	115 589.08	1.1	115 869.80	0.7
2	116 381.73	2.5	116 578.80	2.8	116 800.38	2.2	117 038.77	–4.1	117 312.12	2.2
3	117 811.65	2.7	117 997.38	2.7	118 206.44	2.0	118 437.40	2.1	*118 687.40*	
4	119 178.66	1.5	*119 350.59*		*119 546.18*		*119 761.03*		*119 995.16*	

υ	*N* = 11	O–C	*N* = 12	O–C	*N* = 13	O–C	*N* = 14	O–C	*N* = 15	O–C
0	114 681.89	0.4	115 021.56	0.6	115 381.58	0.6	115 761.35	1.0	116 158.96	1.1
1	116 173.29	1.3	*116 495.64*		*116 838.88*		*117 200.41*		*117 578.91*	

D_2 $^3\Pi_g^-$ <4f>$^-$ s

υ			*N* = 2	O–C	*N* = 3	O–C	*N* = 4	O–C	*N* = 5	O–C
0			118 021.90	0.3	118 136.63	0.3	118 280.16	0.3	118 451.03	–0.3

[a] Observed values in normal type. Calculated values (when no observed is available) are given in *italics* with no O–C values. In one case a clear assignment of the MQDT calculated states is not possible and O–C is given as "—." Terms from assignments made in the present work (Table 4) are given in **bold**. Underlined observed values may be questionable (as indicated by current results). Doubly underlined observed values are based on "*calculated only*" <2c>$^-$ c lower state terms (those in italics in Table 1). Any change to the lower state terms would require the according adjustment made to the doubly underlined observed values. Levels flagged with a * are treated more appropriately in a companion article [5].
The observed values are from:
H_2: except where noted, data are from Dieke [34]. Terms flagged Δ represent changes from Dieke. Note that before use all terms from Dieke were lowered by 149.6 cm^{-1} [36].
HD: all data are from Tables V and VI of Keiding and Bjerre [38], combined with the energy, 97 107.50 cm^{-1}, of the <2c>$^-$ c, υ=0, *N*=1 state from Table VI of Kim and Mazur [39] (as interpreted from their eq. (6), note that their abstract and conclusions appear to misprint this energy value.)
D_2: all data are from Freund, Schiavone, and Crosswhite [12], except those flagged with an "*m*" (which are from Jungen et al. [40]).

[b] States are labelled with Dieke's notation given in triangular brackets, e.g., <3e>, where "*ne*" corresponds to "$nd\pi_g$" and "*nf*" corresponds to "$nd\delta_g$". The Kronig parity is added as a superscript, e.g., <3e>$^-$. The traditional or Herzberg notation, where such exists, is given by the following characters, e.g., <3e>$^-$ i.

[c] Observed–calculated value in cm^{-1}.

[d] Terms from Table I of Jungen et al. [37].
Terms flagged *e–l* were obtained from published transitions by adding the observed transition wavenumbers to the lower state <2c>$^-$ c terms of Table 1, averaging where appropriate. Terms derived from transitions from "*Calculated only*" lower state terms (those in italics in Table 1) were only included in the averaging if no transitions from "observed" lower states were available and are doubly underlined.

[e] Terms obtained from the transitions in Tables I–IV of Józefowski, Ottinger, and Rox [52].

[f] Terms obtained from the transitions of Table I of Koot et al. [31].

[g] Terms obtained from the transitions of Table I of de Bruijn and Helm [7].

[h] Terms obtained from the transitions of Table II of Eyler and Pipkin [32].

[i] Terms obtained from transitions assigned in the present work (see Table 4). These were unassigned transitions in Table II of Eyler and Pipkin [32]. Given in **bold**.

[j] Terms obtained from the transitions of Table III of Lembo et al. [9].

[k] Terms obtained from the transitions of Table I of Siebbeles et al. [33].

[l] Terms obtained from the transitions of Table I of Lembo et al. [8] with the exception of the terms for the υ = 11 and 12 *N* = 3 terms that were obtained by adding the term value for <2c>$^-$ c, υ = 9, *N* = 1 from our Table 1 to the relative terms in their Table IV.

[m] Terms from Table I of Jungen et al. [40].

[υ] υ1, υ2, ... denote calculated levels lying near or above the lowest involved ionization limit. These were obtained with ionic vibrational basis sets starting with υ^+ = 1, 2, …, respectively.

Table 4. Comparison of Eyler and Pipkin's[a] (E&P), and Dieke's[b] assignments of <4e>$^-$ r state levels with MQDT results (cm^{-1}).

H_2 $^3\Pi_g^-$ <4e>$^-$ r

	$N = 1$		$N = 2$				$N = 3$			
υ	Dieke[b]	E&P	Dieke	E&P	RJM[c]	Transition[d]	Dieke	E&P	RJM	Transition
0	–23.9	–0.4	–0.2	–0.3			–0.2	–0.4		
1	–11.0	–0.3	–0.1	0.0			–0.7	–0.1		
2	20.3	–0.3	10.8	–0.1			7.1	–0.1		
3	–271.2		–292.6		0.2	22 100.815	–281.3		0.0	22 099.046

	$N = 4$				$N = 5$				$N = 6$	
υ	Dieke	E&P	RJM	Transition	Dieke	E&P	RJM	Transition	Dieke	E&P
0	–0.4	–0.4			–0.5	–0.4			–0.6	–0.6
1	–0.3	–0.2			–2.3	–0.3			–3.4	–0.3
2	1.6	–0.1			–3.2		–0.2	22 275.302[e]	–52.3	–0.3
3	17.1		0.0	22 092.187			–0.2	22 080.766		

[a] Eyler and Pipkin [32] values minus current MQDT calculated values.
[b] Dieke [34] values minus current MQDT calculated values.
[c] New assignments minus current MQDT calculated values.
[d] New assignments of unassigned transitions in Table II of Eyler and Pipkin [32]. These are $\Delta\upsilon = 0$, R-branch transitions from the <2c>$^-$ c state to the <4e>$^-$ r state.
[e] Eyler and Pipkin [32] give two possibilities for this assignment. The present work indicates that this is the correct one.

We believe that Table 3 represents the most complete and up-to-date compilation of gerade-d terms for H_2, HD, and D_2 available. Many of the levels reported in Table 3 lie well above the ionization threshold and the $n = 2$ dissociation limit (identified in the table for the $n = 3$ states). These levels should most correctly be treated by an open-channel version of MQDT. However, we defer this to a companion article [5], flagging those levels considered therein with a * in Table 3. Other resonances, not reported in Table 3, are known in this region but have not yet been clearly assigned into the Rydberg manifold [8, 9, 41]. Many of these, however, are treated in the companion article.

The calculated energies in the present work have all been obtained in a closed channel formalism and we therefore expect a deterioration of the agreement with experiment for those high-lying levels. This is indeed seen in the results in Table 3. Many of these high-lying levels, in particular those of the <4f>$^-$j state are significantly predissociated with widths as large as 96 cm^{-1} [33]. It is therefore not surprising that the Obs. – Calc. values for some of these levels are rather large.

Exciting by laser from the metastable <2c>$^-$c state, Eyler and Pipkin [32] have reexamined the lower vibrational levels of the <4e>$^-$r state. Their work has resulted in some significant changes to the energy levels of Dieke [34]. This can best be seen by comparing the Obs. – Calc. term values for Dieke's terms and for those terms derived from Eyler and Pipkin's assignments. We show this comparison in Table 4. The changes in term values resulting from the assignments given by Eyler and Pipkin are substantial and in every case considered here are confirmed by the MQDT calculations. Furthermore, on the basis of the present calculations, it is now possible to assign the transitions observed around 22 100 cm^{-1} as $\Delta\upsilon = 0$, R-branch transitions to the $\upsilon = 3$ state — exactly as anticipated by Eyler and Pipkin. These newly assigned transitions are given in the "transition" column of Table 5 and, when combined with the <2c>$^-$c state terms of Table 1, give rise to the new terms in the RJM column.

The single transition to the <4e>$^-$r, $\upsilon = 2$, $N = 5$ level was uncertain in Eyler and Pipkin, there being two candidates separated by approximately 3 cm^{-1}. The quality of the present calculations has allowed us to choose unambiguously which of these is correct.

Table 5. Comparison of current (MQDT) observed–calculated values with those of previous work.

<2c>⁻ c		H_2			HD			D_2		
	υ	Experiment[a]	BO[b]	MQDT	Experiment	BO[b]	MQDT	Experiment	BO[b]	MQDT
$N = 1$	0	94 941.58	15.8	–4.0	95 066.63	11.9	–2.9	95 215.52	7.7	–2.0
	1	97 280.60	13.6	–3.6	97 107.50	10.3	–2.8	96 897.10	7.1	–1.8
	2	99 497.27	11.4	–3.5	99 056.17	8.9	–2.7	98 516.54	6.4	–1.7
	3	101 595.29	9.6	–3.3	100 914.99	7.5	–2.6	100 075.35	5.5	–1.6
	4	103 577.28	7.8	–2.9				101 574.69	4.9	–1.5
	5							103 015.45	4.2	–1.6

<3e>⁻ i		H_2					HD				D_2			
	υ	Experiment[c]	BO[b]	Adiab.[d]	Non-Ad.[e]	MQDT	Experiment	BO[b]	Adiab.[d]	MQDT	Experiment	BO[b]	Adiab.[d]	MQDT
$N = 1$	0	112 066.65	127.7	–0.9	–139.	–0.7	112 178.84	96.3	–0.7	–0.5	112 317.02	64.4	–0.7	0.6
	1	114 179.22	118.6	–1.2	–248.	–0.2	114 027.69	90.3	–1.0	–0.2	113 845.27	61.3	–0.7	1.7
	2	116 145.57	109.5	–2.1	–340.	0.0	115 767.54	84.4	–1.5	0.2	115 301.01	58.1	–0.9	1.5
	3	117 958.86	101.0	–2.9	–405.	–0.1	117 395.20	78.6	–2.1	0.2	116 683.29	54.9	–1.2	1.2
	4	119 596.4			–470.	–0.5					117 975.17	37.5	–15.8	–13.2
$N = 2$	0	112 140.82		–50.2	–130.	–0.8	112 243.26		–22.0	–0.6	112 365.66		–9.7	0.6
	1	114 259.06		–28.2	–243.	–0.2	114 093.72		–16.4	–0.2	113 893.63		–7.6	1.7
	2	116 227.95		–19.9	–336.	0.1	115 833.87		–12.2	0.3	115 348.78		–6.0	1.5
	3	118 041.08		–13.8	–403.	–0.2	117 460.67		–9.2	0.1	116 730.18		–4.8	1.2
	4	119 675.7			–470.	–1.2					118 038.20		–0.8	4.0
$N = 3$	0	112 264.88		–85.2	–127.	–1.0	112 344.04		–49.1	–0.7	112 439.25		–22.3	0.4
	1	114 384.38		–62.1	–239.	–0.4	114 194.55		–37.0	–0.4	113 966.40		–17.4	1.5
	2	116 353.23		–43.7	–332.	–0.2	115 933.87		–27.2	0.1	115 420.39		–13.5	1.5
	3	118 164.91		–28.5	–400.	–0.4	117 559.06		–19.1	0.2	116 800.31		–10.1	1.1
	4	119 797.4			–466.	0.5					118 103.67		–7.1	0.9
$N = 4$	0	112 441.77		–131.6	–128.	–1.1	112 483.10		–79.4	–0.9	112 538.45		–37.3	0.3
	1	114 557.69		–99.0	–239.	–0.6	114 331.52		–60.8	–0.4	114 063.94		–29.4	1.4
	2	116 523.39		–70.5	–331.	–0.1	116 068.28		–45.0	0.0	115 516.05		–22.7	1.4
	3	118 330.71		–45.4	–398.	–0.5	117 690.39		–31.1	–0.1	116 893.90		–16.7	1.1
	4	119 955.			–468.	–1.3					118 194.84		–11.1	0.8

Table 5. (*continued*).

<4e>$^-$ r		H_2					H_2 (continued)					H_2 (continued)		
	υ	Experiment[c]	L-Range[f]	MQDT		υ	Experiment	L-Range[f]	MQDT		υ	Experiment	L-Range[f]	MQDT
$N = 1$	0	117 587.66	7.2	–0.4	$N = 3$	0	117 706.10	9.4	–0.4	$N = 5$	0	118 099.26	10.4	–0.5
	1	119 739.35	10.5	–0.3		1	119 873.14	13.4	–0.1		1	120 246.72	14.8	–0.3
	2	121 756.54	12.1	–0.3		2	121 904.43	17.1	0.0		2	**122 260.82**	**20.5**	**–0.2**
$N = 2$	0	117 599.44	9.0	–0.3	$N = 4$	0	117 874.13	9.9	–0.4					
	1	119 770.24	12.5	0.0		1	120 033.08	13.8	–0.2					
	2	121 802.95	15.9	–0.1		2	122 057.75	18.9	–0.1					

<3f>$^-$ j		H_2				HD			D_2		
	υ	Experiment[c]	BO[g]	Non-Ad.[e]	MQDT	Experiment[c]	BO[h]	MQDT	Experiment[c]	BO[g]	MQDT
$N = 2$	0	112 513.95	88.4	–77.	0.6	112 616.21	57.8	0.4	112 749.45	33.12	0.1
	1	114 706.80	72.3	–162.	1.4	114 536.56	47.9	0.8	114 336.37	29.7	0.3
	2	116 776.23	59.3	–231.	2.3	116 362.68	36.9	2.1	115 859.36	26.5	1.7
	3	118 725.16	48.6	–271.	1.1	118 096.83	22.9	1.4	117 319.74	23.6	2.0
	4	120 555.6	38.4	–284.	2.2				118 718.86	20.9	1.0
	5								120 057.46	17.7	0.6
$N = 3$	0	112 732.18		–89.	1.0				112 849.93		0.3
	1	114 904.61		–175.	1.9				114 430.86		0.6
	2	116 954.76		–242.	2.7				115 948.15		1.9
	3	118 885.31		–280.	1.6				117 403.21		2.1
	4	120 695.8		–292.	4.5				118 796.88		1.1
$N = 4$	0	113 007.64		–101.	1.4				112 981.74		0.4
	1	115 158.20		–188.	2.4				114 555.31		0.8
	2	117 186.46		–254.	3.3				116 065.43		2.1
	3	119 094.20		–290.	2.1				117 513.51		2.3
	4	120 863.4		–317.	–7.3				118 900.06		1.3

Table 5. (*concluded*).

<4f>$^-$ s		H_2				D_2		
	υ	Experiment	BO[g]	L-Range[f]	MQDT	Experiment[c]	BO[g]	MQDT
$N = 2$	0	117 830.35	152.3	9.2	0.1	118 021.90	49.7	0.3
	1	119 997.24	121.7	11.5	0.8			
	2	122 042.65	96.5	18.2	1.2			
	3	123 969.04	75.1	27.4	1.0			
$N = 3$	0	118 067.15		9.1	0.2			
	1	120 226.08		17.9	0.3			
	2	122 246.82		16.1	1.2			
	3	124 155.12		23.5	1.1			
$N = 4$	0	118 352.97		8.9	0.3			
	1	120 491.70		14.2	0.7			
	2	122 493.10		9.0	1.1			
	3	124 387.56		20.9	1.3			

[a] Experimental values from Table 1.
[b] Born–Oppenheimer results derived from Kolos and Rychlewski [44].
[c] Experimental values from Table 3. Underlined term is questionable, **bold** term is assigned in current work.
[d] Adiabatic results from Rychlewski [46].
[e] Non-adiabatic results from Bak and Linderberg [47].
[f] Long-range model results from Eyler and Pipkin [32].
[g] Born–Oppenheimer results derived from Rychlewski [50].
[h] Born–Oppenheimer results derived from Rychlewski [51].

For one highly excited vibrational level seen by Lembo et al. [9] (<4e>$^-$r, $\upsilon = 9$, $N = 4$) it was not possible to clearly identify which of the near-lying bound-channel calculated levels corresponded to the physical level. We have therefore not reported an Obs. – Calc. value for this state.

Eyler and Pipkin also observed transitions to the <4f>$^-$s state and there again their work resulted in significant changes to the term values as compared with the values of Dieke [34]. As was the case for the <4e>$^-$r state their new assignments to the <4f>$^-$s state are all validated by the present calculations.

(Some of these states have been reported in emission in ref. 42. However, discrepancies between the values in the transition and term tables lead us to rely on other experimental data for comparison [3].)

5. Comparison with other calculations

The calculations we have presented are ab initio calculations with no fitting to experimental data. In this section, we compare our MQDT results with those of other ab initio calculations for the same levels. This comparison is detailed in Table 5, which we discuss below. Calculations other than those shown in Table 5 have been performed, for example Schins et al. [43] have carried out a fitted coupled-equations calculation on the $n = 3$ triplet gerade complex. As in the Schins et al. calculation, all others of which we are aware involve adjustments to fit to experimental data and are not strictly comparable with the current fully ab initio results. The reader is reminded that the MQDT treatment *is* amenable to least-squares fitting where appropriate, as was briefly illustrated in ref. 3. In the present case, however, the quality of the ab initio results means that such a fitting was not required.

5.1. $^3\Pi_u^-$ symmetry: <2c>$^-$c state

For the triplet ungerade levels the only previous calculations of which we are aware were done in the Born–Oppenheimer approximation for the lowest state, the $n = 2$, <2c>$^-$c state, by Kołos and Rychlewski [44]. Unfortunately, their results are given in the less than felicitous form of dissociation energies of hypothetical rotationless $N = \Lambda = 0$ levels. To compare their ab initio results with experimental values in the most straightforward manner we have decreased the dissociation energies from their Table VII by $1 \times B_\upsilon$ (for $[N(N+1) - \Lambda^2]B_\upsilon$), using rotational constants, B_υ, determined from the expectation values for R^{-2} they report in the same table. These adjusted dissociation energies we then subtracted from the ab initio dissociation limits of Dressler and Wolniewicz [45] to arrive at term values comparable to experiment. The resulting Obs. – Calc. values are given in the B–O column of the <2c>$^-$c state section of Table 5. The Born–Oppenheimer results are quite reasonable for this lowest state. (Note, however, that in their Table X Kołos and Rychlewski make a different comparison with experiment and report that their band origin lies 125.92 cm^{-1} too low.) To the best of our knowledge, there have been no ab initio calculations of higher lying rovibronic levels of $^3\Pi_u^-$ symmetry.

5.2. $^3\Pi_g^-$ and $^3\Delta_g^-$ symmetry

In Table 5 of ref. 3, we reported earlier MQDT calculations on the lowest gerade states of d-symmetry, the <3e>$^-$i and <3f>$^-$j states. The results of our previous calculations for these states were of similar quality to these obtained here. One new aspect of the current work is the incorporation of the energy dependence of the relevant quantum defects, now allowing us to continue our calculations not only to higher vibrational levels of these states but also to higher levels of the Rydberg series. The other new aspect is the incorporation of the doubly-excited channels for the $^3\Pi_g^-$ symmetry, which allows us, in a companion article [5], to account for the dissociation pathway through the long-range, doubly excited portion of the <3e>$^-$i state. That we are able to account for the strong interaction between the singly excited and doubly excited channels without degrading the quality of the rovibronic results for the bound states must be considered a further verification of the flexibility of MQDT.

5.2.1. $\langle 3e\rangle^-i$ state

A variety of previous calculations have been performed for this lowest of the $^3\Pi_g^-$ states. In the same article in which they presented the results for the $\langle 2c\rangle^-$c state Kołos and Rychlewski [44] also presented Born–Oppenheimer results for the $\langle 3e\rangle^-$i state. We have converted the dissociation energies in their Table VIII to absolute energies in the same way as we did for their $\langle 2c\rangle^-$c levels above, taking the expectation values for R^{-2} from their Table VIII and the dissociation energies from ref. 45. The resulting Obs. – Calc. values are given in the B–O column of our Table 5. Note that whereas we find an Obs. – Calc. value of 127.7 cm^{-1} for their lowest calculated level, in their Table X Kołos and Rychlewski report a disagreement with experiment of 237.87 cm^{-1}. In either case the Born–Oppenheimer approximation can be seen to have broken down even for this first $d\pi_g$ state.

More recently Rychlewski [46] has calculated the electronic adiabatic corrections for the $\langle 3e\rangle^-$i state and calculated the rovibrational levels of this state taking them into account. His results are given in his Table 4 and reproduced here in the Adiab. column of Table 5. (Note that Rychlewski's Obs. – Calc. values are somewhat different due to the fact that he compares his calculated values with the problematic experimental results of ref. 42.) For $N = 1$ it can be seen that the adiabatic corrections result in a stunning improvement in the calculated energy levels. The MQDT results are also seen to completely account for the adiabatic effects. For $N > 1$ the adiabatic results can be seen to deteriorate quickly and dramatically. This is due to the fact that for $N > 1$ the $\langle 3e\rangle^-$i levels mix with the $\langle 3f\rangle^-$j levels, an effect not accounted for by the simple adiabatic picture.

Prior to the adiabatic calculations of Rychlewski, Bak and Linderberg [47] performed nonadiabatic calculations of both c- and d-symmetry levels of the $n = 3$ triplet gerade manifold of states in 1990. To bring their results into better agreement with experiment they seem to have shifted all their calculated levels down by approximately 645 cm^{-1} to bring the $\langle 3a\rangle$h, $\upsilon = 0$, $N = 0$ level into exact agreement with experiment. The resulting shifted energy levels give the Obs. – Calc. values presented as Non-adiab. in Table 5 for the $\langle 3e\rangle^-$i and $\langle 3f\rangle^-$j states. The agreement with experiment is not particularly striking.

In principle, since they are only the first elements of their respective Rydberg series, accurate nonadiabatic calculations of levels of the $\langle 3e\rangle^-$i state and the $\langle 3f\rangle^-$j state must be possible. Dressler and co-workers have performed such calculations for the analogous states in the singlet manifold, first including five coupled electronic states in 1990 [48] then including nine states in 1994 [49], and obtained excellent agreement with experiment for the first elements of the Rydberg series.

Examining Table 5, one sees that the present MQDT results account as precisely for the adiabatic and nonadiabatic interactions in the $\langle 3e\rangle^-$i state as did the calculations in ref. 3 in 1994.

5.2.2. $\langle 4e\rangle^-r$ state

The $n = 4$ state of $^3\Pi_g^-$ symmetry was given by Dieke [34]. As discussed above, Eyler and Pipkin [32] have reexamined this state and given substantial reassignments, which have been confirmed in their entirety by the present calculations. Eyler and Pipkin also made use of a long-range polarizability model incorporating quadrupole interactions between the Rydberg electron and the H_2^+ ion core. The quality of this simple ab initio model aided their task in assigning the spectrum. We report their Obs. – Calc. values in the L-range column of Table 5. Once again the quality of the MQDT results is immediately apparent in this table.

5.2.3. $\langle 3f\rangle^-j$ state

As for $\langle 2c\rangle^-$c and $\langle 3e\rangle^-$i states, Rychlewski [50] presents Born–Oppenheimer results for the $\langle 3f\rangle^-$j state for the H_2 isotopomers, with results for HD available from his earlier work [51]. The same conversion of their dissociation energies for hypothetical rotationless states to actual term values comparable with experiment has been done as we did for the $\langle 2c\rangle^-$c and $\langle 3e\rangle^-$i states, with the exception that the rotational correction is now $2 \times B_\upsilon$ (for $[N(N+1) - \Lambda^2]\ B_\upsilon$) and the B_υ values are taken from Rychlewski's earlier work [51], while the dissociation energies are, as always, from ref. 45. From the outset, the Obs. – Calc. values are significantly different from zero, due to the fact that all of the levels of this $^3\Delta_g^-$ state are mixed with the levels of the $^3\Pi_g^-$ $<3e>^-$i state, an effect not accounted for in the Born–Oppenheimer or nonadiabatic approximations. (Note that Rychlewski's own presentation of his disagreement with experiment for H 12 is over 100 cm^{-1} (Table V of ref. 51).)

Bak and Linderberg's nonadiabatic calculations [47] also included the $<3f>^-$j state for H_2. Their results are reported in Table 5 and are at best only qualitative.

6. $<$4f$>^-$s state

Results of Born–Oppenheimer calculations for the $n = 4$ state of $^3\Delta_g^-$ symmetry are given in Table IV of the paper by Rychlewski [50]. In this case, dissociation energies for actual $N = \Lambda = 2$ levels are given in Table IV of ref. 50. Combining these with the appropriate dissociation energies from ref. 45 we obtain the Obs. – Calc. values given in the B–O column, which in this particular case agree with that reported in Table VI by Rychlewski himself . Yet again the Born–Oppenheimer results are in very poor agreement with experiment.

Unsurprisingly, the long-range polarizability-quadrupole model used by Eyler and Pipkin [32] is in considerably better agreement with experiment than the Born–Oppenheimer results, while the MQDT results are in even better agreement thereby fully confirming the major reassessment of this state that Eyler and Pipkin have given.

Thus, in each case considered, MQDT has provided the best, and often by far the best, results for the rovibronic energy levels of these states. This is remarkable when it is remembered that all of these calculations have been done using the same unified MQDT framework and using only two sets of quantum defect functions (one set for ungerade symmetry, the other for gerade symmetry), both sets having been extracted from the same clamped-nuclei electronic potential-energy curves as were used in the much less successful Born–Oppenheimer calculations.

7. Conclusions

The initial motivation of the present work was to prepare the ground for our treatment of the continuum processes presented in a companion paper [5]. Nevertheless several new elements have been incorporated such as the $\tilde{\mu}$-defect, the inclusion of more than one doubly excited state, and a more systematic treatment of energy dependence. We have stressed that the strong R-dependence of the $^3\Pi_g$ quantum defect matrix elements displayed in Fig. 2*a* does not in itself preclude a good description of the non-Born–Oppenheimer effects. Indeed, the quality of the results shown in the tables presented here matches the best one might anticipate in any case. Given this quality, we plan to apply the present theoretical framework to other symmetries and to identify higher rovibronic levels in the spectrum of the hydrogen molecule and its isotopomers, notably based on the Dieke tables [34] in which the majority of the transitions remain unassigned. We also plan to use MQDT to account for the hyperfine structure resolved in the recent set of experiments by Ottinger and collaborators [52, 53].

In a companion paper [5], we shall see that the quantum defects presented here also give a satisfactory account of the fragmentation/ionization dynamics at higher energy. Taken together this pair of articles illustrates the unifying power of MQDT with a single set of R-dependent body-frame quantities, the quantum defects, successfully accounting for nonadiabatic interactions in both the bound-state manifold and in the continuum, along with the effects of competition between ionization and dissociation in the continuum.

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